# Supplementary results: affine companions, spreads and exact-order statistics

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**Abstract**

We give coordinate Singer constructions, classifications of fixed complete mappings through oriented spreads, a nonlinear companion pair generating $A_{16}$, and further scalar exact-order results. These results accompany *Norm-fiber densities and relative traces in characteristic two*. We also give the detailed parameter comparisons with prescribed-order predecessors and the computational and formalization record. Numbers with an $S$ prefix refer to this supplement; unprefixed references refer to the main paper.

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# S1 The six-dimensional coordinate example

For the six-bit coordinate example, $V = \mathbb{F}_2^6$; other dimensions are stated explicitly. Coordinates $x = (x_1, \ldots, x_6)$ have $x_1$ the most significant bit (value 32) and $x_6$ the least (value 1); a word is identified with its integer value in $\{0, \ldots, 63\}$. Basis $G = (\mathtt{63}, \mathtt{33}, \mathtt{18}, \mathtt{49}, \mathtt{1}, \mathtt{21})$ (a fixed ordered basis — the six words are linearly independent over $\mathbb{F}_2$, by Gaussian elimination; non-uniqueness is discussed in §S2); write $g_j$ for the $j$-th entry, $j = 0, \ldots, 5$. $\mathrm{coords}_G(x) = (c_1, \ldots, c_6)$ is the unique vector with $x = \bigoplus_{i=1}^{6} c_i\, g_{i-1}$, read as the six-bit word $c_1 \cdots c_6$ ($c_1$ most significant); $\oplus$ is bitwise XOR, i.e. addition in $\mathbb{F}_2^n$. When a field structure is chosen in characteristic two, $\oplus$ is also its field addition. A monic polynomial is called *primitive* if it is irreducible and its roots generate the multiplicative group of its root field. Superscripts on functions and on $F$ denote composition iterates; orders of field elements denote multiplicative orders.

## S1.1 An affine Singer map

### Definition and basic checks

$T(x) = 63 \oplus \mathrm{coords}_G(x)$. Verified identities: $T(0) = 63$, $T(41) = 0$, $T(38) = 38$; for example $\mathrm{coords}_G(38) = (0, 1, 1, 0, 0, 1)$, read as $011001 = 25$, so $T(38) = 63 \oplus 25 = 38$, and $\mathrm{coords}_G(41) = (1, 1, 1, 1, 1, 1) = 63$, so $T(41) = 63 \oplus 63 = 0$. The integer labels $38, 41, 63$ belong to this chosen coordinate presentation; they are not invariants under changes of basis (see §S2).

**Lemma S1.1.** *$T$ is affine: $T(x) = L_T x + b$ with $b = T(0) = 63$ and $L_T(x) = T(x) \oplus 63$.*

*Proof.* Since $G$ is a basis, $\mathrm{coords}_G$ is an $\mathbb{F}_2$-linear isomorphism. Thus $L_T = \mathrm{coords}_G$, and adding the constant 63 gives the stated affine map. □

**Lemma S1.2.** $\mathrm{Fix}(T) = \{38\}$*; on the remaining* 63 *points $T$ is a single cycle (type* $1 + 63$*, order* 63*), equivalently $L_T$ has order* 63 *(Proposition S1.3).*

**Proposition S1.3.** *Identify $(V, +)$ with $\mathbb{F}_{64}$; then $L_T$ is multiplication by a primitive element of $\mathbb{F}_{64}$, a Singer cycle [3]. The characteristic polynomial of $L_T$ is*

$$p_T(t) = t^6 + t^5 + t^3 + t^2 + 1, \tag{S1.1}$$

*computed directly as* $\det(tI + L_T)$ *(code appendix), primitive of degree* 6*.*

*Proof.* The matrix in Appendix S.B gives the displayed determinant. Modulo $p_T$, repeated squaring and multiplication give

$$t^{64} = t, \qquad \gcd(p_T, t^8 + t) = \gcd(p_T, t^4 + t) = 1.$$

The degree-6 irreducibility criterion, using the prime divisors $2, 3$ of 6, therefore shows that $p_T$ is irreducible. For completeness, $p_T(0) = p_T(1) = 1$ also gives $\gcd(p_T, t^2 + t) = 1$. In $\mathbb{F}_2[t]/(p_T)$ the residue $\bar{t}$ satisfies

$$\bar{t}^{63} = 1, \qquad \bar{t}^{21} = \bar{t}^4 + \bar{t}^2 + \bar{t} + 1 \neq 1, \qquad \bar{t}^9 = \bar{t}^2 + \bar{t} \neq 1.$$

Since $63 = 3^2 \cdot 7$, its order is 63. The irreducible characteristic polynomial identifies $L_T$ with multiplication by $\bar{t}$ on this field. Finally, $T(38) = 38$ implies $T(x) \oplus 38 = L_T(x \oplus 38)$, proving the fixed-point and cycle claims of Lemma S1.2. □

**Powers and** $63 = 7 \cdot 9 = 3 \cdot 21$

Since $L_T$ has order 63, $T^d$ for $d \mid 63$ has type $1 + d \times (63/d)$ (one fixed point plus $d$ cycles of length $63/d$): $T^1 = 1 + 63$, $T^3 = 1 + 3 \cdot 21$, $T^7 = 1 + 7 \cdot 9$, $T^9 = 1 + 9 \cdot 7$, $T^{21} = 1 + 21 \cdot 3$ (verified).

## S1.2 The complete-mapping triple

A permutation $f$ of an additive group of exponent two is a *complete mapping* if $\mathrm{id} \oplus f$ is again a permutation. Define its *triality operator* $F(f) = (\mathrm{id} \oplus f)^{-1}$, where $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$. By Proposition S1.3, identify the linear part of $T$ with multiplication by a primitive $\alpha \neq 1$. Then $\mathrm{id} \oplus T$ has nonzero slope $1 \oplus \alpha$ and is therefore a permutation. Its inverse $D = F(T)$ has slope $(1 \oplus \alpha)^{-1}$, and $1 \oplus (1 \oplus \alpha)^{-1} = \alpha/(1 \oplus \alpha) \neq 0$, so $\mathrm{id} \oplus D$ is also a permutation. Thus $D = F(T) = (\mathrm{id} \oplus T)^{-1}$ and $K = F(D) = (\mathrm{id} \oplus D)^{-1}$ are defined; since Lemma 2.2 gives $F^3(T) = T$, $\{T, D, K\}$ is the $F$-orbit of $T$. Verified facts:

- cycle types $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$ (seven 9-cycles plus one fixed point);
- fixed points $\mathrm{Fix}(T) = \{38\}$, $\mathrm{Fix}(D) = \{41\}$, $\mathrm{Fix}(K) = \{63\}$;
- $F(K) = T$, i.e. $(\mathrm{id} \oplus K)^{-1} = T$; hence $F^3 = \mathrm{id}$ with $T \to D \to K \to T$;
- offset-fixed three-cycle: $D(0) = 38 = \mathrm{Fix}(T)$, $K(0) = 41 = \mathrm{Fix}(D)$, $T(0) = 63 = \mathrm{Fix}(K)$;
- linear parts: for any $n \times n$ matrix $A$ in characteristic two with $p_A(t) = \det(tI + A)$ and $p_A(1) \neq 0$, taking determinants gives

$$p_{(I+A)^{-1}}(t) = \frac{\det\big(t(I + A) + I\big)}{\det(I + A)} = \frac{t^n p_A(1 + t^{-1})}{p_A(1)}.$$

  Here $p_T(1) = 1$ and $p_T(t + 1) = p_T(t)$ (directly, $(t + 1)^6 + (t + 1)^5 + (t + 1)^3 + (t + 1)^2 + 1$ reduces to $t^6 + t^5 + t^3 + t^2 + 1$). Hence $L_D = (I + L_T)^{-1}$ has $p_D(t) = t^6 p_T(1 + t^{-1}) = t^6 p_T(t^{-1}) = t^6 + t^4 + t^3 + t + 1$ (primitive). The reciprocal form uses this shift invariance. $L_K$ has $\Phi_9(t) = t^6 + t^3 + 1$ (cyclotomic, non-primitive, order 9).

For the Singer example, in field coordinates $T(x) = \alpha x \oplus c$, so $F(T)(0) = D(0) = (1 \oplus \alpha)^{-1} c$ solves $x = \alpha x \oplus c$, i.e. $D(0) = \mathrm{Fix}(T)$, and the slope cycle $\alpha \mapsto (1 \oplus \alpha)^{-1} \mapsto 1 \oplus \alpha^{-1} \mapsto \alpha$ gives the two remaining identities. For the six-bit layer the values are $F(T)(0) = 38$, $F(D)(0) = 41$, $F(K)(0) = 63$. Conceptually, $T$ is the launching map — an affine Singer layer read through the fixed generator basis (a change of basis) — while the triality operator $F$ keeps the third-order cycle running: $T \mapsto D \mapsto K \mapsto T$, carrying the fixed point $38 \mapsto 41 \mapsto 63 \mapsto 38$.

**Theorem S1.4** (The six-dimensional Singer application)**.** $\langle T, D \rangle = \mathrm{AGL}(1, 64)$*; its order is* $4032 = 64 \cdot 63$.

*Proof.* By Proposition S1.3, in field coordinates the linear part of $T$ is multiplication by a primitive $\alpha \in \mathbb{F}_{64}$. The translation is $c = T(0) = 63 \neq 0$. Thus $U = V$ and the least common multiple in Corollary 2.4 is already 63 because $\mathrm{ord}(\alpha) = 63$. The corollary proves the claim. □

The general structure allows reducible and nonsemisimple matrices.

### The exponent convention

For the six-bit example, the fixed point 38 can serve as the origin after relabeling, while 0 remains the additive origin. In the value set $\{0, \dots, 63\}$ the 9-multiples are exactly the 8 self-doubled words $(x_i = x_{i+3})$, a 3-dimensional subspace. To parametrize the long orbit, define

$$\theta(j) = T^j(63), \qquad j \in \mathbb{Z}/63.$$

This is a bijection onto $V \setminus \{38\}$. Apply the CRT isomorphism to the *iteration exponent* $j$, sending it to $(j \bmod 7, j \bmod 9)$. Thus $\theta(0) = 63$ corresponds to $(0, 0)$, whereas $0 = \theta(62)$ corresponds to $(6, 8)$, since $T(0) = 63$. Reduction of the integer word labels modulo 63 does not give this parametrization. The cyclotomic factor $\Phi_{2^k+1}$ reflects $(2^k - 1)(2^k + 1) = 2^{2k} - 1$; the fixed point of $K$ accounts for the remaining point.

## S2 Uniform statement for $2k$-bit words

Identify the $2k$-dimensional $\mathbb{F}_2$-space $V_{2k}$ with $\mathbb{F}_{2^{2k}}$; write $M = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$. In field coordinates an affine Singer layer is a map $T(x) = \alpha x \oplus b$ whose linear part is multiplication by a primitive element $\alpha$ (order $M$).

**Lemma S2.1.** *For every $k \geq 1$, $\mathrm{ord}_{2^k+1}(2) = 2k$.*

*Proof.* Let $m = \mathrm{ord}_{2^k+1}(2)$. Since $2^k \equiv -1 \pmod{2^k + 1}$, we have $m \mid 2k$, while $m \nmid k$. Suppose, for contradiction, that $m < 2k$. If $m$ were odd, $m \mid 2k$ would imply $m \mid k$. Thus $m$ is even; write $m = 2d$ with $d \mid k$, and $m < 2k$ gives $d < k$. The subgroup $\langle 2 \rangle \subset (\mathbb{Z}/(2^k+1)\mathbb{Z})^\times$ is cyclic of order $2d$, so its unique element of order 2 is $2^d$. But $2^k \equiv -1$ also has order 2 in this subgroup; hence $2^d \equiv -1 \pmod{2^k + 1}$. Thus $2^k + 1 \mid 2^d + 1$, contradicting $0 < 2^d + 1 < 2^k + 1$. Therefore $m = 2k$. □

For $q = 2^k$, define

$$\mathrm{PTE}(k): \quad \exists \beta \in \mathbb{F}_{q^2}^*, \quad \mathrm{ord}(\beta) = q^2 - 1, \quad \mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = 1.$$

The following theorem supplies the required existence input; its proof is given in Appendix S.A.

**Theorem S2.2** (Classical PTE existence)**.** *For every $k \geq 1$, $\mathrm{PTE}(k)$ holds.*

*Proof.* See Appendix S.A. □

**Theorem S2.3.** *For every $k \geq 1$ there exists a primitive $\alpha \in \mathbb{F}_{2^{2k}}$ such that, for every $b \in \mathbb{F}_{2^{2k}}^*$, the affine Singer layer $T_b(x) = \alpha x \oplus b$ has $D_b = (\mathrm{id} \oplus T_b)^{-1}$ and $K_b = (\mathrm{id} \oplus D_b)^{-1}$ defined. The maps $T_b, D_b$ have cycle type $1 + M$, while $K_b$ has cycle type $1 + (2^k - 1) \times (2^k + 1)$, and $\langle T_b, D_b \rangle = \mathrm{AGL}(1, 2^{2k})$. The triality operator satisfies $F^3 = \mathrm{id}$ with $T_b \mapsto D_b \mapsto K_b \mapsto T_b$.*

*Proof.* Let $b \neq 0$ be arbitrary and write $T = T_b$, $D = D_b$, $K = K_b$. The linear parts of $T$, $D$, $K$ are $\alpha$, $(1 \oplus \alpha)^{-1}$, and $\kappa = 1 \oplus \alpha^{-1} = (1 \oplus \alpha)/\alpha$ respectively (independent of the translation $b$); the three slopes are cycled by the map $z \mapsto 1/(z \oplus 1)$, so $F^3 = \mathrm{id}$. Since $\mathbb{F}_{2^{2k}}^*$ is cyclic, an affine map with slope $\mu \neq 1$ has cycle type $1 + \frac{M}{\mathrm{ord}(\mu)} \times \mathrm{ord}(\mu)$ (the translation only moves the fixed point; a shorter orbit would force $T^d$, $d < \mathrm{ord}(\mu)$, to have a second fixed point, impossible since $\mu^d \neq 1$); the stated types therefore follow once $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = M$ and $\mathrm{ord}(\kappa) = 2^k + 1$. Existence of such an $\alpha$ follows directly from Theorem S2.2. Put $q = 2^k$ and choose a primitive $\beta \in \mathbb{F}_{q^2}$ with $\beta^q \oplus \beta = 1$. Take $\alpha = \beta$. Then $1 \oplus \alpha = \alpha^q$, so

$$\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = q^2 - 1 = M, \qquad \kappa = \frac{1 \oplus \alpha}{\alpha} = \alpha^{q-1}, \qquad \mathrm{ord}(\kappa) = q + 1.$$

Here the last equality follows from $\mathrm{ord}(\alpha^{q-1}) = M/\gcd(M, q-1) = q+1$. These are exactly the required slope orders. To make the dependence on $b$ explicit, set $S_b(x) = bx$. Then $T_b = S_b T_1 S_b^{-1}$. Indeed, for any invertible additive map $S$ and $g = SfS^{-1}$,

$$(\mathrm{id} \oplus g)(x) = S\big((\mathrm{id} \oplus f)(S^{-1}x)\big).$$

Taking inverses gives $F(g) = SF(f)S^{-1}$ whenever $F(f)$ is defined. Applying this with $S = S_b$ gives $D_b = S_b D_1 S_b^{-1}$ and $K_b = S_b K_1 S_b^{-1}$. Their fixed points are

$$\mathrm{Fix}(T_b) = \{b/(1 \oplus \alpha)\}, \qquad \mathrm{Fix}(D_b) = \{b/\alpha\}, \qquad \mathrm{Fix}(K_b) = \{b\}.$$

Thus all nonzero $b$ give the stated cycle types; §2 gives the group-generation claim. □

Since the $K$-slope has order $2^k+1$, the degree over $\mathbb{F}_2$ of its minimal polynomial is $\mathrm{ord}_{2^k+1}(2) = 2k$ by Lemma S2.1; this polynomial is therefore an irreducible degree-$2k$ factor of $\Phi_{2^k+1}$. **Remark (the $\tau$-collapse).** Put $q = 2^k$. For $\gamma \in \mathbb{F}_{q^2}^*$ of order $q+1$, one has $\gamma^q = \gamma^{-1}$; hence $\tau = \gamma \oplus \gamma^{-1}$ satisfies $\tau^q = \tau$ and belongs to $\mathbb{F}_q$. The Frobenius identities $(1 \oplus \gamma)^{2^k-1} = \gamma^{-1}$ and $(1 \oplus \gamma)^{2^k+1} = \tau$, together with $\gcd(2^k-1, 2^k+1) = 1$, give the following order calculation. For an element $z$ of finite order $N$ and coprime positive integers $r, s$, $\mathrm{ord}(z^r) = N/\gcd(N, r)$ implies $N = \mathrm{lcm}(\mathrm{ord}(z^r), \mathrm{ord}(z^s))$. Consequently $\mathrm{ord}(1 \oplus \gamma) = \mathrm{lcm}(\mathrm{ord}(\gamma^{-1}), \mathrm{ord}(\tau)) = \mathrm{lcm}(2^k+1, 2^k-1) = M$ whenever $\tau$ is primitive in $\mathbb{F}_{2^k}$; the pair $(u^{2^k-1}, u^{2^k+1}) = (\gamma^{-1}, \tau^{-1})$ for $u = \gamma/(1 \oplus \gamma)$ gives the same order for $u$. Since $\alpha = (1 \oplus \gamma)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1} = \gamma$, once $\gamma$ has order $2^k+1$ the three order conditions on $\alpha$ reduce to the single condition that $\tau$ be primitive. The order assumption on $\gamma$ is essential; it is not implied by $\tau$ being primitive alone. Existence of such a $\gamma$ is the classical square-order statement (see [19] and [20, Theorem 1, p. 1]; for an equivalent square-order formulation, see [18, Theorem 1.5, p. 170]); it is also supplied directly by Theorem S2.2. For $q = 2^k$ the appendix proof is self-contained, so the paper does not use these classical results as external existence inputs.

For this coordinate application, an ordered generator basis is called *admissible for the stated Singer triple* when the resulting linear slopes satisfy $\mathrm{ord}(\alpha) = \mathrm{ord}(1+\alpha) = 2^{2k}-1$ and $\mathrm{ord}(1+\alpha^{-1}) = 2^k+1$. These order conditions specify exactly which coordinate presentations realize the theorem's triple. The argument of §2 applies verbatim with $Q = 2^{2k}$ (taking $b \neq 0$ in the proof), so the generation $\langle T, D\rangle = \mathrm{AGL}(1, 2^{2k})$ holds for every $k$. For the concrete coordinate presentation of §S1.1 the orders and cycle data were verified for $k = 1, 2, 3, 4$ (the bases are listed in the appendix): $k = 1$ (2 bits) order $12 = 4 \cdot 3$, $K$ type $1+1 \cdot 3$, factor $\Phi_3$ (basis $(1, 3)$: $\mathrm{Fix}(T) = \{1\}$, $\mathrm{Fix}(D) = \{2\}$, $\mathrm{Fix}(K) = \{3\}$); $k = 2$ (4 bits) order $240 = 16 \cdot 15$, $K$ type $1 + 3 \cdot 5$, factor $\Phi_5$; $k = 3$ (6 bits) order $4032 = 64 \cdot 63$, $K$ type $1 + 7 \cdot 9$, factor $\Phi_9$ (this paper); $k = 4$ (8 bits) order $65280 = 256 \cdot 255$, $K$ type $1 + 15 \cdot 17$, linear part an irreducible degree-8 factor of $\Phi_{17}$.

The basis is not canonical even in dimension 6: among the 720 orderings of these six generators, 572 do not even make $\mathrm{id} \oplus T$ a permutation; of the 148 that do, only 54 give $T$ of order 63, and only 9 give the exact cycle-type triple $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$. For instance $(4\,6)$ keeps $T = 1 + 63$ but changes $\mathrm{Fix}(T)$ from 38 to 11 and the cycle types of $D$ and $K$ to $D = 1 + 3 \cdot 21$, $K = 1 + 63$. Hence the particular entries of $G$ and the anchors $\{38, 41, 63\}$ are coordinates of one representative, not invariants, and no uniqueness statement is made. The identity $F^3 = \mathrm{id}$ holds for every complete mapping. The group order and the specified cycle triple follow for the admissible presentations defined above; completeness alone does not impose those three slope orders. The general formula of Theorem 2.3 applies to every presentation, and determines its actual group.

**Remark (constructiveness).**
Theorem S2.3 is an existence statement; its existence input is proved in Appendix S.A. The $\tau$-collapse turns it into a finite search with certification. For $\tau = \gamma \oplus \gamma^{-1} \in \mathbb{F}_{2^k}$, the Artin–Schreier identity

($x^2 + \tau x + 1$ *is irreducible over* $\mathbb{F}_{2^k}$ *iff* $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(1/\tau) = 1$) gives $2^{k-1}$ candidate values of $\tau$. For a candidate, fix an explicit representation of $\mathbb{F}_{2^{2k}}$ and take a root $\gamma$ of $x^2 + \tau x + 1$ (found, for instance, by exhaustive search); the collapse gives the required orders for $1 \oplus \gamma$ and $\gamma/(1 \oplus \gamma)$ precisely when $\tau$ is primitive *and* $\mathrm{ord}(\gamma) = 2^k + 1$. The second condition is not automatic from the first: for example, when $k = 6$ and $\mathbb{F}_{64} = \mathbb{F}_2[t]/(t^6 + t + 1)$, the element $\tau = t^5 + t^3 + 1$ is primitive and $\mathrm{Tr}(1/\tau) = 1$, but the roots of $x^2 + \tau x + 1$ have order 13, not 65. The theorem of Appendix S.A guarantees that a jointly successful candidate exists, so a deterministic enumeration that tests both conditions terminates. Certifying them requires factoring $2^k - 1$ and $2^k + 1$ (or an equivalent exact-order test). This procedure supplies a deterministic construction with certification; no factorization-free closed formula is supplied here. A necessary restatement is that $u = 1/\tau$ is a primitive element of $\mathbb{F}_{2^k}$ with $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(u) = 1$; equivalently, its degree-$k$ minimal polynomial has $x^{k-1}$-coefficient 1. This condition alone is still not sufficient for admissibility, because the root-order condition above must also be checked.

## S3 Fixed mappings and orbits under $F$

Lemma 2.2 implies that every orbit under $F$ has length one or three. We first give the scalar classification, then describe all fixed complete mappings geometrically and count the orbits in the entire affine class. An *$F$-fixed map* means $F(f) = f$; this is distinct from the unique fixed point of the permutation $f$ in Lemma 2.1.

### S3.1 Coordinate-permutation symmetry

**Proposition S3.1.** *For a complete mapping $f$ of an additive group $V$ of exponent two, put*

$$\Gamma_f = \{(x, f(x), x \oplus f(x)) : x \in V\}, \qquad J(f) = f^{-1}.$$

*Permuting the three coordinates preserves this family of sets. The permutations $\rho(x, y, z) = (z, x, y)$ and $\sigma(x, y, z) = (y, x, z)$ induce $F$ and $J$, respectively. Thus $J$ preserves complete mappings and*

$$F^3 = J^2 = \mathrm{id}, \qquad JFJ = F^{-1} = F^2.$$

*For finite $V$, $\Gamma_f$ is a transversal of its additive Cayley table.*

*Proof.* Every triple of $\Gamma_f$ has sum zero, and each of its three coordinate projections is a bijection to $V$. Both properties are invariant under coordinate permutations. Conversely these properties recover a unique complete mapping from the first two coordinates. With $z = x \oplus f(x)$, one has $F(f)(z) = x$ and $z \oplus x = f(x)$, so

$$\rho(\Gamma_f) = \Gamma_{F(f)}.$$

With $y = f(x)$, swapping the first two coordinates gives $(y, f^{-1}(y), y \oplus f^{-1}(y))$, hence $\sigma(\Gamma_f) = \Gamma_{J(f)}$. The coordinate permutations satisfy $\rho^3 = \sigma^2 = \mathrm{id}$ and $\sigma\rho\sigma = \rho^{-1}$; injectivity of $f \mapsto \Gamma_f$ transfers these relations to $F, J$. For finite $V$, the three bijective projections say exactly that the set contains one cell in each row, each column, and each symbol class of the additive Cayley table. This is the definition of a transversal. □

This is the cyclic part of the classical coordinate-permutation symmetry of Latin squares [1], acting here on transversals of a table invariant under all six coordinate permutations. The proof identifies this relationship explicitly; the order-three symmetry itself is not claimed as new.

### Scalar affine maps

**Proposition S3.2.** *Let $k \geq 1$, $Q = 2^{2k}$, and $\mathcal{A}_Q = \{f_{a,c} : x \mapsto ax \oplus c \mid a \in \mathbb{F}_Q \setminus \{0,1\},\ c \in \mathbb{F}_Q\}$. The operator $F$ acts on $\mathcal{A}_Q$ with two singleton orbits and $(Q(Q-2)-2)/3$ orbits of length three. Its fixed maps are exactly $f_{a,0}$ with $a^2+a+1=0$. On the invariant subset $c \neq 0$, every orbit has length three.*

*Proof.* In coefficient pairs,

$$F(a,c) = \left((1+a)^{-1}, \frac{c}{1+a}\right).$$

The new slope again lies outside $\{0,1\}$, and Lemma 2.2 gives $F^3 = \mathrm{id}$, so each orbit has length one or three. A fixed pair must satisfy $a = (1+a)^{-1}$, equivalently $a^2+a+1=0$; the constant-term equation is $c(1+a) = c$, so $ac = 0$ and $c = 0$. Conversely these conditions give a fixed pair. The polynomial $a^2+a+1$ has two distinct roots in $\mathbb{F}_Q$, since $\mathbb{F}_4 \subseteq \mathbb{F}_Q$. There are $Q(Q-2)$ pairs in total, yielding the stated counts. Multiplication of $c$ by $(1+a)^{-1} \neq 0$ preserves $c \neq 0$, and no fixed pair lies in that subset. □

### Fixed maps and line spreads

A *line spread* of $V = \mathbb{F}_2^n$ is a collection of two-dimensional $\mathbb{F}_2$-subspaces whose nonzero vectors partition $V \setminus \{0\}$. Thus a line here is two-dimensional as a vector space; it is a projective line in $\mathrm{PG}(n-1,2)$. Each such subspace has exactly three nonzero vectors $x, y, x \oplus y$. A cyclic orientation on a line specifies one of the two 3-cycles on them.

**Proposition S3.3.** *For $n \geq 1$, the $F$-fixed complete mappings of $\mathbb{F}_2^n$ are in bijection with line spreads equipped with a cyclic orientation on each line. Every such map fixes 0 and has only 3-cycles on the nonzero vectors. They exist exactly when $n$ is even. If $s_n$ denotes the number of line spreads in a fixed coordinate space, their number, for even $n$, is*

$$s_n \, 2^{(2^n-1)/3}.$$

*Nonlinear $F$-fixed complete mappings exist for every even $n \geq 4$.*

*Proof.* If $F(f) = f$, then $f^{-1} = \mathrm{id} \oplus f$. Evaluating at $f(x)$ gives

$$f^2(x) = x \oplus f(x) = f^{-1}(x), \qquad f^3 = \mathrm{id}.$$

Let $p$ be the unique fixed point from Lemma 2.1. The displayed equality at $p$ gives $p = p \oplus p = 0$. Thus each nonzero cycle is $(x, f(x), x \oplus f(x))$, the nonzero part of a two-dimensional subspace. Disjoint cycles supply a line spread and their orientations. Conversely, orient each line of a spread and fix 0. The resulting permutation satisfies $f^2 = \mathrm{id} \oplus f = f^{-1}$, so it is complete and $F(f) = f$. This proves the bijection and the count.

A spread requires $3 \mid 2^n - 1$, hence even $n$. If $n = 2m$, identify $V$ with $\mathbb{F}_4^m$; its one-dimensional $\mathbb{F}_4$-subspaces give a line spread over $\mathbb{F}_2$, proving existence. For $m \geq 2$ (equivalently, even $n \geq 4$), start with the linear map $A : x \mapsto \omega x$, where $\omega^2 + \omega + 1 = 0$ in $\mathbb{F}_4$, and reverse the orientation on exactly one spread line $U$. The resulting map $f$ equals $A^2$ on $U \setminus \{0\}$ and equals $A$ elsewhere, and remains $F$-fixed. For $0 \neq u \in U$ and $v \notin U$, one has $u \oplus v \notin U$, and

$$f(u \oplus v) = Au \oplus Av \neq A^2 u \oplus Av = f(u) \oplus f(v),$$

since $(A^2 + A)u = u \neq 0$. Hence $f$ is nonlinear. □

If $V$ is identified with $\mathbb{F}_{2^n}$ for even $n$, the spread obtained from $\mathbb{F}_4$ consists of the multiplicative cosets of $\mathbb{F}_4^*$. An orientation on each coset is multiplication by either $\omega$ or $\omega^2$. Thus these examples are a particular family of the first-order cyclotomic complete mappings considered in [22]. Proposition S3.3 describes fixed maps from arbitrary line spreads, without choosing a field structure or restricting to this coset partition.

For $n = 4$, exact-cover enumeration of the 35 lines on 15 nonzero vectors gives $s_4 = 56$, recovering the classical spread count also formalized in Coq by Magaud [2]. The proposition then gives $56 \cdot 2^5 = 1792$ fixed complete mappings. Enumeration of their orientations finds 112 linear and 1680 nonlinear maps, in agreement with the affine count below. These finite counts, the complete list of spreads, and a nonlinear witness are reproduced by `verify_complete_mapping_orbits.py`.

### All affine complete mappings

Write $h_j = |\mathrm{GL}(j,2)| = \prod_{i=0}^{j-1}(2^j - 2^i)$, where $h_0 = 1$ is the empty product, and let

$$\mathcal{C}_n^{\mathrm{aff}} = \{x \mapsto Lx \oplus c : \ L, I + L \in \mathrm{GL}(n,2),\ c \in \mathbb{F}_2^n\}.$$

This includes matrices that are not scalar multiplication in a single field structure on $\mathbb{F}_2^n$.

**Proposition S3.4.** *For $n \geq 1$, put*

$$d_n = h_n \sum_{j=0}^{n} \frac{(-1)^j 2^{j(j-1)/2}}{h_j}, \qquad r_n = \begin{cases} 0, & n \text{ odd}, \\ h_n/|\mathrm{GL}(n/2,4)|, & n \text{ even}. \end{cases}$$

*There are $2^n d_n$ affine complete mappings. Under $F$ they form $r_n$ singleton orbits and $(2^n d_n - r_n)/3$ orbits of length three. The fixed maps are exactly $x \mapsto Lx$ with $L^2 + L + I = 0$.*

*Proof.* The number $d_n$ counts invertible matrices with no nonzero fixed vector. Its displayed expression is the zero-dimensional fixed-space case of the classical Rudvalis–Shinoda count, recorded in [24, Theorem 2.1]. Here is a direct proof. For a $j$-dimensional subspace $U$ put $\mu(U) = (-1)^j 2^{j(j-1)/2}$. If $W$ has dimension $r$, the Gaussian-binomial identity gives

$$\sum_{U \leq W} \mu(U) = \sum_{j=0}^{r} \binom{r}{j}_2 (-1)^j 2^{j(j-1)/2} = \prod_{i=0}^{r-1}(1 - 2^i).$$

This is 1 when $W = 0$ and 0 otherwise. Apply it to $W = \ker(I + L)$ and sum over $L \in \mathrm{GL}(n,2)$. For a fixed $j$-subspace $U$, the matrices fixing $U$ pointwise have block form $\left(\begin{smallmatrix} I & B \\ 0 & C \end{smallmatrix}\right)$, with $2^{j(n-j)}$ choices of $B$ and $h_{n-j}$ choices of $C$. Since there are $\binom{n}{j}_2$ choices of $U$ and

$$\binom{n}{j}_2 2^{j(n-j)} h_{n-j} = h_n / h_j,$$

interchanging the two finite sums yields the formula for $d_n$. Each such $L$ permits all $2^n$ translations $c$.

For $f(x) = Lx \oplus c$, the condition $F(f) = f$ is equivalent to $(\mathrm{id} \oplus f) \circ f = \mathrm{id}$. Comparing linear and constant parts gives $L^2 + L = I$ and $Lc = 0$, so $c = 0$. The polynomial $z^2 + z + 1$ is irreducible over $\mathbb{F}_2$. Thus such an $L$ equips $V$ with an $\mathbb{F}_4$-vector-space structure, which requires even $n$. Conversely, for even $n$ multiplication by a root of this polynomial supplies such an $L$. All these matrices form one conjugacy class in $\mathrm{GL}(n,2)$: choose an $\mathbb{F}_4$-basis in each structure and compare the resulting $\mathbb{F}_2$-bases. The invertible maps commuting with $L$ are exactly the $\mathbb{F}_4$-linear ones, so the centralizer has order $|\mathrm{GL}(n/2,4)|$. This proves the formula for $r_n$. Finally Lemma 2.2 restricts the other orbits to length three. □

The following small values are exact evaluations of the formulas, independently checked by enumerating all binary matrices and translations in these dimensions:

| $n$ | $d_n$ | $\lvert\mathcal{C}_n^{\mathrm{aff}}\rvert$ | $F$-fixed maps | 3-element orbits |
|---|---|---|---|---|
| 2 | 2 | 8 | 2 | 2 |
| 3 | 48 | 384 | 0 | 128 |
| 4 | 5824 | 93184 | 112 | 31024 |

Every $F$-fixed affine map is linear. In particular the 1680 nonlinear fixed maps in dimension four lie outside the affine class, showing the additional scope of Proposition S3.3.

### Linear conjugacy of oriented spreads

We now distinguish the $F$-orbits of maps from equivalence under a linear change of coordinates. Write $f \sim g$ if $g = SfS^{-1}$ for some $S \in \mathrm{GL}(n,2)$. Such conjugacy preserves $F(f) = f$. For a spread $\mathcal{S}$, let $G_{\mathcal{S}}$ be its setwise stabilizer in $\mathrm{GL}(n,2)$, and choose a reference cyclic orientation on each line. An orientation is then a vector $(\epsilon_L)_{L\in\mathcal{S}}$ over $\mathbb{F}_2$, with 1 meaning reversal of the reference orientation. For $g \in G_{\mathcal{S}}$ define $\delta_g(L) \in \mathbb{F}_2$ by comparing the transported reference orientation on $gL$ with the chosen reference on $gL$. Thus the transported bits satisfy

$$\epsilon'_{gL} = \epsilon_L \oplus \delta_g(L).$$

**Proposition S3.5** (Orientation-sensitive orbit formula)**.** *The correspondence of Proposition S3.3 identifies linear conjugacy with equivalence of oriented spreads. Let $c(g)$ be the number of cycles in the permutation of spread lines induced by $g$, and put*

$$e(g) = \begin{cases} 1, & \bigoplus_{L\in C} \delta_g(L) = 0 \text{ for every line cycle } C, \\ 0, & \text{otherwise.} \end{cases}$$

*The number of linear-conjugacy classes supported on the orbit of $\mathcal{S}$ is*

$$a(\mathcal{S}) = \frac{1}{|G_{\mathcal{S}}|} \sum_{g\in G_{\mathcal{S}}} e(g) 2^{c(g)}.$$

*Summing this over representatives of the linear equivalence classes of underlying spreads gives all $F$-fixed linear-conjugacy classes. The stabilizer of a fixed map is the stabilizer of its oriented spread.*

*Proof.* Linear conjugacy sends each cycle $(x, f(x), x \oplus f(x))$ to its transported cycle, so it sends the spread and its orientations to exactly those of $SfS^{-1}$. To fix an orientation, the displayed bit relation must hold with $\epsilon' = \epsilon$. Around a line cycle the successive equations are consistent if and only if the sum of the $\delta_g(L)$ on that cycle is zero. When consistent, there are two choices for its first bit and all other bits are forced. Thus $g$ fixes $e(g)2^{c(g)}$ orientations. Burnside's orbit-counting lemma gives the formula. Restricting to the stabilizer of one underlying spread suffices for its full linear orbit; distinct underlying spread orbits cannot give conjugate maps. The same transport correspondence gives the stabilizer assertion. □

**Proposition S3.6** (Local criterion for an invariant orientation)**.** *Let $\mathcal{S}$ be any line spread and $G = G_{\mathcal{S}}$ its setwise stabilizer in $\mathrm{GL}(V)$. For $L \in \mathcal{S}$, write $G_L = \{g \in G : gL = L\}$. The following conditions are equivalent:*

1. *There is an orientation of every line in $\mathcal{S}$ that is fixed by all of $G$.*

2. *Every $g \in G_L$, for every $L \in \mathcal{S}$, induces an even permutation of the three nonzero vectors of $L$.*

3. *The orientation-translation cocycle associated with any choice of reference orientations is a coboundary.*

*More precisely, put $E = \mathbb{F}_2^{\mathcal{S}}$ with its natural permutation $G$-module structure $(g \cdot v)_{gL} = v_L$. If $D_g \in E$ is defined by $(D_g)_{gL} = \delta_g(L)$, then $D_{gh} = D_g + g \cdot D_h$, and condition (3) means that there exists $v \in E$ with $D_g = v + g \cdot v$ for every $g \in G$.*

*Proof.* An orientation of a line is a cyclic ordering of its three nonzero vectors, up to cyclic rotation. Its stabilizer within the permutation group on those vectors is the alternating group of degree three. Hence (1) implies (2).

Assume (2). Choose one representative $L$ from each $G$-orbit on $\mathcal{S}$ and choose an arbitrary cyclic orientation on it. Transport the chosen orientation to $gL$ using $g \in G$. If $gL = hL$, then $h^{-1}g \in G_L$ and (2) says that this element preserves the chosen orientation. Thus the transport is independent of the choice of $g$, and the resulting orientation on all lines is fixed by $G$. This proves (2)⇒(1).

For the last equivalence, the transported bit vector is $g \cdot v + D_g$. Composing two transports gives $D_{gh} = D_g + g \cdot D_h$. The bit vector $v$ is fixed by every $g$ precisely when $v = g \cdot v + D_g$, or equivalently $D_g = v + g \cdot v$. This is exactly the coboundary condition. □

The proposition distinguishes the existence of one orientation fixed by the entire spread automorphism group from the individual fixed-orientation tests $e(g)$ in Proposition S3.5; the latter still supply the general orbit count even when no globally invariant orientation exists.

**Theorem S3.7** (Four-dimensional classification)**.** *There are exactly three conjugacy classes under* $\mathrm{GL}(4,2)$ *of $F$-fixed complete mappings of $\mathbb{F}_2^4$. Identify $V$ with $\mathbb{F}_4^2$, choose $\omega^2 + \omega + 1 = 0$, and start with $f_0(x) = \omega x$ on its five spread lines. Representatives are obtained by reversing any $r$ lines, with $r = 0, 1, 2$. Their orbit sizes, stabilizer orders and stabilizer groups are*

| *r* | *orbit size* | *stabilizer order* | *stabilizer group* | *linear* |
|---|---|---|---|---|
| *0* | *112* | *180* | $C_3 \times A_5$ | *yes* |
| *1* | *560* | *36* | $C_3 \times A_4$ | *no* |
| *2* | *1120* | *18* | $C_3 \times S_3$ | *no* |

*Here $C_3$ is cyclic of order three, and $A_j, S_j$ denote alternating and symmetric groups. In particular the* 1680 *nonlinear maps form two classes.*

*Proof.* First every four-dimensional spread is equivalent to the displayed one. Choose two spread lines as complementary spaces $E \oplus E$, where $E = \mathbb{F}_2^2$. The other three lines are graphs of invertible maps $P_i : E \to E$, with all $P_i + P_j$ invertible. A change of coordinates makes one of these maps $I$. Among the six matrices of $\mathrm{GL}(2,2)$, the only ones for which $I + P$ is invertible are

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

They satisfy $P^2 + P + I = 0$. The three graphs are consequently those of $I, P, P^2$, giving the five one-dimensional $\mathbb{F}_4$-subspaces of $\mathbb{F}_4^2$.

For this standard spread, the only linear maps inducing cyclic orientations on its lines are $f_0$ and $f_0^2$. Indeed on each line such a map is multiplication by $\omega$ or $\omega^2$; additivity at $e_1 + e_2$ forces the choices on the two coordinate lines to agree, which determines the map on their direct sum. Therefore any linear stabilizer of the underlying spread conjugates $f_0$ to either $f_0$ or $f_0^2$. The commuting

maps are $\mathrm{GL}(2,4)$; composing with the coordinatewise Frobenius map $\sigma(x_1,x_2)=(x_1^2,x_2^2)$ accounts for the other coset. Thus the full spread stabilizer is $\Gamma\mathrm{L}(2,4)$ of order $2|\mathrm{GL}(2,4)|=360$. Since $|\mathrm{GL}(4,2)|=20160$, there are $20160/360=56$ spreads, recovering the classical count [2].

On the five lines, $\mathrm{GL}(2,4)$ has scalar kernel $C_3$ and image $\mathrm{PGL}(2,4)$ of order 60. This index-two subgroup of $S_5$ is $A_5$. Its elements preserve every reference orientation. Frobenius reverses them all and induces the transposition exchanging $\omega$ and $\omega^2$ on $\mathbb{P}^1(\mathbb{F}_4)$, fixing $0,1,\infty$. Consequently even line permutations permute the reversed-line subset, whereas odd line permutations also complement it. $A_5$ is transitive on subsets of each fixed size: an arbitrary permutation carrying one subset to another can have its parity adjusted by a transposition inside the subset or its complement. Hence the orientation orbits are exactly the weight pairs $\{0,5\}$, $\{1,4\}$, $\{2,3\}$, of sizes $2,10,20$. Multiplying by the 56 underlying spreads gives the three displayed orbit sizes and orbit–stabilizer gives their stabilizer orders.

For the representatives of weights $r=0,1,2$, an odd line permutation cannot stabilize the orientation because it changes the weight to $5-r$. Within $\mathrm{GL}(2,4)$ the projective stabilizers are $A_5$, $A_4$, and the two-subset stabilizer $S_3$, respectively. The determinant splits $\mathrm{GL}(2,4)=C_3\times\mathrm{SL}(2,4)$; the projective action identifies $\mathrm{SL}(2,4)$ with $A_5$, giving the displayed direct products. The earlier two-linear-map observation shows that only the first class is linear. □

The proof gives representatives and stabilizers, rather than only the total number of maps. Independently, `verify_affine_groups_and_conjugacy.py` constructs all 1792 maps and conjugates the three representatives by all 20160 matrices; their disjoint orbits exhaust the list. It also compares direct fixed-orientation counts with Proposition S3.5 for all 360 standard-spread automorphisms. For dimension six, underlying spreads have already been classified by Mateva and Topalova [25] into 131044 types. We next treat the standard Desarguesian type, one of these types. The other underlying types are outside the six-dimensional computation reported here.

### Affine group orders in dimension four

| group order | 3 | 12 | 15 | 48 | 192 | 240 |
|---|---|---|---|---|---|---|
| number of maps | 112 | 3360 | 4032 | 5040 | 20160 | 60480 |

Of the 93184 affine complete mappings, 80640 give transitive groups; 60480 give primitive groups, all doubly transitive in this dimension. These counts are computational evaluations of the structure theorem, reproduced by the additional verifier in Appendix S.B; the action criteria are proved in main Corollary 2.4 for every dimension.

### Standard spreads in arbitrary even dimension

Let $V=\mathbb{F}_4^m$, let $\omega^2+\omega+1=0$, and let $\mathcal{S}_m$ consist of its one-dimensional $\mathbb{F}_4$-subspaces, viewed as two-dimensional $\mathbb{F}_2$-subspaces. Put

$$N_m=\frac{4^m-1}{3},\qquad P_m=\mathrm{PGL}(m,4),\qquad p_m=|P_m|=\frac{1}{3}\prod_{i=0}^{m-1}(4^m-4^i).$$

Orient each line by $x\mapsto\omega x$ before choosing the reversed-line subset. Write $a_m$ for the number of full linear-conjugacy classes supported on spreads equivalent to $\mathcal{S}_m$.

**Theorem S3.8** (Standard-spread action and asymptotics)**.** *For $m\geq 2$, the setwise stabilizer of $\mathcal{S}_m$ in $\mathrm{GL}(2m,2)$ is $\Gamma\mathrm{L}(m,4)$. Its linear coset permutes orientation bits; its semilinear coset also complements every bit and fixes no orientation. Every oriented stabilizer contains the scalar group*

*$C_3 = \mathbb{F}_4^*$ and is contained in* $\mathrm{GL}(m,4)$. *If $c(g)$ counts the cycles of $g$ on the $N_m$ projective points, then*

$$a_m = \frac{1}{2p_m} \sum_{g \in P_m} 2^{c(g)} = \frac{2^{N_m}}{2p_m}(1+\varepsilon_m), \qquad 0 \le \varepsilon_m \le (p_m-1)2^{-2^{2m-3}}. \tag{S3.1}$$

*For a uniformly chosen orientation,*

$$\Pr(\mathrm{Stab}_{\mathrm{GL}(2m,2)}(f) \neq C_3) \le (p_m-1)2^{-2^{2m-3}}. \tag{S3.2}$$

*Consequently almost every orientation has stabilizer exactly $C_3$ as $m \to \infty$, and*

$$a_m \sim \frac{3\,2^{(4^m-1)/3}}{2|\mathrm{GL}(m,4)|}.$$

*Such orientations exist for every $m \ge 4$ and are nonlinear. Furthermore, exactly $a_m - 1$ of the $a_m$ supported conjugacy classes are nonlinear for every $m \ge 2$.*

*Proof.* A linear map giving cyclic orientations on the lines of $\mathcal{S}_m$ must be either $f_0(x) = \omega x$ or $f_0^2(x) = \omega^2 x$. On each line it has one of these two restrictions. Applying additivity to independent coordinate vectors and their sum forces their choices to agree, and the two $\mathbb{F}_2$-basis vectors on each coordinate line then determine the map on $V$. A spread stabilizer conjugates $f_0$ to one of these two maps. The centralizer of $f_0$ is $\mathrm{GL}(m,4)$; coordinatewise Frobenius $\sigma(x) = x^2$ conjugates $f_0$ to $f_0^2$. This proves the spread-stabilizer assertion.

Linear elements preserve the reference orientations; elements of $\mathrm{GL}(m,4)\sigma$ reverse all of them. Such an element sends an orientation of weight $r$ to one of weight $N_m - r$. As $N_m$ is odd, it fixes none. Scalars act trivially on the orientation set, giving the first equality in (S3.1) by Proposition S3.5. The orientation stabilizer is therefore the inverse image in $\mathrm{GL}(m,4)$ of its subset stabilizer in $P_m$.

For a nonscalar matrix $A$, its fixed projective points are the lines in its eigenspaces for the three nonzero elements of $\mathbb{F}_4$. If their dimensions are $d_\lambda$, the number is $\sum_\lambda (4^{d_\lambda}-1)/3$. Their dimensions sum to at most $m$ and each is at most $m-1$. Convexity gives the upper bound $(4^{m-1}-1)/3+1$. Thus every nonidentity projective element moves at least $4^{m-1}-1$ points. Each nontrivial permutation cycle has at least two points, so

$$N_m - c(g) \ge \left\lceil \frac{4^{m-1}-1}{2} \right\rceil = 2^{2m-3}.$$

It fixes a proportion $2^{c(g)-N_m}$ of all orientations. Summing these proportions bounds both the nonidentity contribution in (S3.1) and, by the union bound, the probability in (S3.2). This is the classical motion argument [26] in the present projective action.

Finally $p_m < 2^{2m^2}/3$ and $2^{2m-3} \ge 2m^2$ for $m \ge 4$; the latter holds at $m = 4$ and increases thereafter. The exceptional bound is then less than one and tends to zero. The only two linear orientations have stabilizer $\mathrm{GL}(m,4)$, which is larger than $C_3$, so orientations with exactly $C_3$ are nonlinear. The two uniform orientations are conjugate by the semilinear Frobenius map, and no other orientation is linear; hence they constitute the single linear conjugacy class and the other $a_m - 1$ classes are nonlinear. □

For example $p_4 = 987033600$, and the exceptional proportion is at most $987033599/2^{32} < 0.230$. The estimate therefore already guarantees that more than 77% of the orientations on the standard eight-dimensional spread have exactly the scalar stabilizer. This conclusion concerns the standard spread family, without asserting a distribution over all spreads.

**Corollary S3.9** (Six-dimensional standard-spread classes)**.** *The* $2^{21}$ *orientations of* $\mathcal{S}_3$ *give exactly* 80 *full* $\mathrm{GL}(6,2)$ *conjugacy classes supported on its underlying spread orbit. Classifying by the smaller of the two complementary orientation weights, the numbers are*

| $r$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *classes* | 1 | 1 | 1 | 2 | 3 | 5 | 8 | 10 | 13 | 17 | 19. |

*Only the weight-zero class is linear.*

*Proof.* Here $|P_3| = 60480$. Its projective cycle types and multiplicities are listed below; a superscript is the number of cycles of that length.

| matrix description | projective cycle type | multiplicity |
|---|---|---|
| scalar | $1^{21}$ | 1 |
| $J_2(a) \oplus a$ | $1^5\,2^8$ | 315 |
| $J_3(a)$ | $1\,2^2\,4^4$ | 3780 |
| $aI_2 \oplus b$, $a \neq b$ | $1^6\,3^5$ | 672 |
| $J_2(a) \oplus b$, $a \neq b$ | $1^2\,2^2\,3\,6^2$ | 10080 |
| three distinct eigenvalues | $1^3\,3^6$ | 2240 |
| quadratic block, normalized order 5 | $1\,5^4$ | 8064 |
| quadratic block, normalized order 15 | $1\,5\,15$ | 16128 |
| cubic block, projective order 3 | $3^7$ | 1920 |
| cubic block, projective order 7 | $7^3$ | 5760 |
| cubic block, projective order 21 | 21 | 11520 |

For completeness these multiplicities follow directly from rational canonical form. In $\mathrm{GL}(3,4)$, the first six types have respectively $3, 3, 3, 6, 6, 1$ conjugacy classes. Their corresponding centralizer orders are

$$181440, \quad 576, \quad 48, \quad 540, \quad 36, \quad 27.$$

Divide their total element counts by three to pass to projective matrices. For a quadratic block plus eigenvalue $a$, divide the quadratic roots by $a$. The 12 elements of $\mathbb{F}_{16} \setminus \mathbb{F}_4$ have orders 5 or 15, with 4 or 8 such roots. They give 6 or 12 matrix classes with centralizer order 45. For a cubic block, roots outside $\mathbb{F}_4$ have projective orders $3, 7, 21$, represented by $6, 18, 36$ field elements. They give $2, 6, 12$ matrix classes with centralizer order 63. For the Jordan types, $J_2(a) \oplus a$ has projective order two and five fixed points. The powers of $J_3(a)$ have respectively one, five, and twenty-one fixed points at exponents $1, 2, 4$, giving $1\,2^2\,4^4$. For $J_2(a) \oplus b$, the corresponding counts at exponents $1, 2, 3, 6$ are $2, 6, 5, 21$, giving $1^2\,2^2\,3\,6^2$. The remaining cycle types follow from the orders on the invariant line, quadratic component, or cubic field. All multiplicities sum to 60480, so the types are exhaustive.

Summing multiplicity times $2^{\text{number of cycles}}$ gives 9676800; (S3.1) gives $9676800/120960 = 80$. The number of $r$-subset classes under $P_3$ is the coefficient of $z^r$ in

$$\frac{1}{60480} \sum_{g \in P_3} \prod_{C \text{ a point cycle of } g} (1 + z^{|C|}).$$

Using the table gives the displayed row. Semilinear conjugation pairs weights $r$ and $21 - r$, so each entry also counts the oriented classes of the corresponding weight pair. Only the two uniform orientations are linear, and they belong to the same class. □

An independent complete traversal of the $2^{21}$ orientation vectors supplies all 80 representatives, their full linear stabilizer orders, and their orbit sizes. Among these, four classes have stabilizer $C_3$, accounting for 483840 orientations on the fixed standard spread. These representative and stabilizer data are finite computations, distinct from the general proof of Theorem S3.8. The supplied verifier

also independently enumerates the 60480 projective matrices and recovers the preceding cycle table and weight polynomial.

The general fixed-map description reduces counting to the line-spread enumeration problem; no general formula for $s_n$ is supplied here. The affine orbit count is complete for every dimension, while the scalar Singer triples provide a distinguished family with controlled permutation cycle types and two-generator affine groups. Theorems 2.3, S3.7 and S3.8 give the general affine group structure, the full four-dimensional classification, and the standard-spread asymptotics. Corollary S3.9 counts all six-dimensional standard-spread oriented classes; it does not classify orientations on the other 131043 underlying spread types (out of 131044 in total).

## S4 Nonlinear companion groups

An $F$-fixed map generates only a cyclic group of order three with its companion. Large companion groups therefore require a three-element $F$-orbit. If two companions both have cycle type $1+(2^n-1)$, their cycles have odd length, so their group is contained in $A_{2^n}$. The following example attains this upper bound in dimension four.

**Proposition S4.1** (Two long cycles with distinct fixed points)**.** *Let $\Omega$ be a finite set of size $N \geq 4$, and suppose $f, g \in \mathrm{Sym}(\Omega)$ both have cycle type $1+(N-1)$. If their unique fixed points are distinct, then $H = \langle f, g \rangle$ acts doubly transitively on $\Omega$. If in addition $N = 2^n$ with $n \geq 3$ and $H$ contains a 3-cycle, then $H = A_N$.*

*Proof.* Let $a$ be the unique fixed point of $f$. Since $g$ fixes a different point, $g(a) \neq a$. The $(N-1)$-cycle $f$ acts transitively on $\Omega \setminus \{a\}$, so $H$ is transitive on $\Omega$: its orbit containing $a$ also contains $g(a)$ and every other point. Moreover $f \in H_a$, and $f$ itself is transitive on $\Omega \setminus \{a\}$. Thus the point stabilizer is transitive on the remaining points and $H$ is doubly transitive.

Now let $N = 2^n \geq 8$. An $(N-1)$-cycle is even, so $H \leq A_N$. The action is primitive since it is doubly transitive. The classical Jordan theorem for a primitive degree-$N$ group containing a prime cycle of length $p \leq N-3$, applied with $p = 3$, implies $A_N \leq H$ (see, e.g., [17]). Therefore $H = A_N$. □

This sufficient condition separates the group-theoretic deduction from the difficult existence problem for complete mappings with prescribed companion cycle types: the distinct-fixed-point hypothesis gives double transitivity automatically, while the existence of a suitable short-cycle element still requires additional arguments.

**Proposition S4.2** (A nonlinear alternating-group example)**.** *On $V = \mathbb{F}_2^4$, with integer labels and XOR addition, let $f$ be the permutation in the first row below and let $g = F(f)$.*

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $f(x)$ | 1 | 5 | 12 | 3 | 11 | 7 | 15 | 10 | 2 | 14 | 9 | 13 | 0 | 8 | 6 | 4 |
| $g(x)$ | 3 | 0 | 5 | 10 | 1 | 13 | 11 | 9 | 14 | 6 | 8 | 15 | 12 | 7 | 2 | 4 |

*Then $f$ is complete and nonlinear. Each of $f, F(f), F^2(f)$ has cycle type $1+15$, with fixed points $3, 12, 1$ respectively, and*

$$\langle f, F(f) \rangle = A_{16}.$$

*Proof.* The first row and $x \oplus f(x)$ each permute the labels $0, \ldots, 15$; inverting the latter gives the

second row. The cycles other than the indicated fixed points are

$$f : (0\ 1\ 5\ 7\ 10\ 9\ 14\ 6\ 15\ 4\ 11\ 13\ 8\ 2\ 12),$$
$$g : (0\ 3\ 10\ 8\ 14\ 2\ 5\ 13\ 7\ 9\ 6\ 11\ 15\ 4\ 1),$$
$$F^2(f) : (0\ 12\ 14\ 7\ 2\ 10\ 13\ 6\ 8\ 5\ 4\ 11\ 15\ 9\ 3).$$

The third cycle follows from $F^2(f) = \mathrm{id} \oplus f^{-1}$. An affine map would satisfy $f(1 \oplus 2) = f(1) \oplus f(2) \oplus f(0)$. Here these values are 3 and 8, so $f$ is nonlinear. Since $F$ preserves the affine class, all three companions are nonlinear.

Use right-to-left composition, write $[u, v] = uvu^{-1}v^{-1}$, and put $w = f^2 g$. Directly from the table,

$$w = (0\ 3\ 14)(1\ 5\ 2\ 10\ 12)(4\ 7\ 6\ 8\ 15\ 13\ 9), \qquad w(11) = 11.$$

Thus $s = w^{35} = (3\ 0\ 14)$. All generators are even, so $H = \langle f, g\rangle \le A_{16}$. The cycles $f^j s f^{-j} = (3\ f^j(0)\ f^j(14))$ belong to $H$, and their supports contain every point because $f$ cycles all points other than 3.

To see that these cycles generate $A_{16}$, make a graph on the other 15 points, joining $a, b$ whenever $(3\ a\ b)$ is available. There are no isolated vertices. If two graph components have edges $ab$ and $cd$, the identity $[(3\ a\ b), (3\ c\ d)] = (3\ a\ c)$ joins them. After the graph is connected, conjugating $(3\ b\ c)$ by $(3\ a\ b)$ gives $(a\ 3\ c)$, so paths supply $(3\ a\ b)$ for every pair. Finally $(3\ a\ b)(3\ b\ c) = (a\ b\ c)$. These are all three-cycles, which generate $A_{16}$. Hence $H \ge A_{16}$, proving equality without a group-order search. □

Alternatively, the distinct fixed points 3 and 12 give double transitivity by Proposition S4.1; the explicit word $w^{35}$ is a 3-cycle. The Jordan-theorem conclusion in that proposition therefore yields $A_{16}$ directly. The elementary proof above is retained independently of Jordan's theorem.

Complete mapping polynomials over $\mathbb{F}_{16}$ and their reduced-degree distribution were computationally studied by Yuan–Tong–Zhang [28]; Bors–Wang studied cycle types and coset-wise affine constructions of complete mappings [22, 27]. These works provide important precedents for the finite-field maps and cycle-type methods. The result here concerns the additional group generated by one specified map and its prescribed companion, which is not the object counted by those cited results. A construction in infinitely many dimensions with simultaneous cycle-type and companion-group control is not supplied here. The groups generated by the full collections of complete mappings in [23] concern a different set of generators.

**Proposition S4.3** (Translations of fixed maps)**.** *Let $f_0$ be an $F$-fixed complete mapping on $\mathbb{F}_2^n$ and let $b \neq 0$. Write $\tau_b(x) = x \oplus b$ and $f = \tau_b f_0$. Then $f$ is complete and*

$$F(f) = f_0\tau_b, \qquad F^2(f) = \tau_b f_0 \tau_b, \qquad \langle f, F(f)\rangle = \langle f_0, \tau_b\rangle.$$

*The orbit of $0$ under this group is exactly the four-point line*

$$U_b = \{0, b, f_0(b), f_0^2(b)\}.$$

*The induced action on $U_b$ is $A_4$. Thus this construction cannot give a transitive companion group when $n > 2$.*

*Proof.* The fixed-map identities give $f_0^3 = \mathrm{id}$ and $\mathrm{id} \oplus f_0 = f_0^2$. Therefore $\mathrm{id} \oplus f = \tau_b f_0^2$ is a permutation, and its inverse is $g = f_0\tau_b$. Also

$$(\mathrm{id} \oplus g)(x) = x \oplus f_0(x \oplus b) = b \oplus f_0^2(x \oplus b),$$

so taking its inverse gives the formula for $F^2(f)$. Since $gf = f_0^2$, the group generated by $f, g$ contains $f_0$ and then $\tau_b = f f_0^{-1}$; the reverse inclusion is immediate.

Proposition S3.3 makes $U_b$ a two-dimensional subspace, invariant under both $f_0$ and $\tau_b$. Starting at 0, translation reaches $b$ and $f_0$ reaches the other two nonzero points, so the orbit is all of $U_b$. On these points $f_0 = (b\ f_0(b)\ f_0^2(b))$ and $\tau_b = (0\ b)(f_0(b)\ f_0^2(b))$. These generate the three double transpositions and a three-cycle, hence $A_4$. □

# S5 Additional scalar consequences

## S5.1 The exact index-three threshold

**Corollary S5.1** (The exact threshold at index three)**.** *For even $n \geq 4$, the specified subgroup*

$$\mathbb{F}_{2^n}^{+} \rtimes C_{(2^n-1)/3}$$

*is primitive and is generated by a scalar affine complete mapping and its prescribed companion if and only if $n \geq 8$. Thus index-three proper primitive companion groups exist in every even dimension from eight onwards.*

*Proof.* For every even $n \geq 4$, writing $t = 2^{n/2} > 2$ gives $(t^2-1)/3 > t-1$. The field-degree argument in Theorem 6.2 proves primitivity. At $n = 4$ the complement has order $5 = 2^2 + 1$, excluded by Theorem 6.5. At $n = 6$ it has order 21, and the polynomial identities following Corollary 2.6 give $\mathcal{R}_{21} = \varnothing$.

For $k = 3$, $b_3 = 2/9$. Using $\omega(d) < 2n/3$, the uniform lower bound in (6.3) is greater than

$$\frac{2^n}{12} - \frac{8}{9} - \left(\frac{2n}{3} + \frac{2}{9}\right) 2^{n/2}.$$

This expression equals $424/3 > 0$ at $n = 14$. After division by $2^{n/2}$, increasing $n$ by two doubles its positive exponential term, whereas its linear term increases by $4/3$ and its negative exponential term decreases in magnitude. Its increase is greater than $2^{n/2}/12 - 4/3 > 0$ for even $n \geq 14$, proving positivity there. For $n = 10$, $d = 341 = 11 \cdot 31$ and $\omega(d) = 2$; for $n = 12$, $d = 1365 = 3 \cdot 5 \cdot 7 \cdot 13$ and $\omega(d) = 4$. The bound $L_{q,3,d}$ equals $40/3$ and $632/9$, respectively, so these dimensions are covered as well.

Finally take $n = 8$ and

$$\mathbb{F}_{256} = \mathbb{F}_2[t]/(m(t)), \qquad m(t) = t^8 + t^4 + t^3 + t + 1, \qquad a = t^5 + t^3 + t^2 + t.$$

The irreducibility criterion for $m$ is certified by $t^{256} \equiv t \pmod{m}$ and $\gcd(m, t^{16} + t) = 1$; these are polynomial identities over $\mathbb{F}_2$. Direct reduction modulo $m$ gives

$$a^{85} = (1+a)^{85} = 1, \qquad a^5 = t^7 + t^5 + t^3 + t + 1 \neq 1, \qquad a^{17} = t^7 + t^6 + t^5 + t^3 + t^2 + 1 \neq 1.$$

Since $85 = 5 \cdot 17$, these identities show that $a$ has exact order 85 and $1 + a \in C_{85}$. Thus $a \in \mathcal{R}_{85}$; $f(x) = ax + c$ realizes the group for every $c \neq 0$ by Corollary 2.5. This closes all the even dimensions claimed. □

The square-root estimate in main Proposition 6.1 is classical; the results specialize it to the subgroup generated by the linked slopes $a$ and $1 + a$. The sufficient bound is not a general necessary condition for realizability, and no priority claim is made for the character-sum method or these particular applications.

## S5.2 Primitive classification and generating triangles

**Proposition S5.2** (Primitive inverse classification and linear conjugacy)**.** *Up to* $\mathrm{GL}(n,2)$ *conjugacy, the primitive groups generated by an affine complete mapping and its prescribed companion are exactly*

$$H_d = \mathbb{F}_{2^n}^+ \rtimes C_d, \qquad d \mid 2^n - 1, \quad \mathrm{ord}_d(2) = n, \quad \mathcal{R}_d \neq \varnothing.$$

*Distinct admissible* $d$ *give distinct group-conjugacy classes. For each such* $d$*, the normalizer in* $\mathrm{GL}(n,2)$ *has order* $n(2^n-1)$ *and is the semilinear scalar group* $\Gamma\mathrm{L}(1,2^n)$*. The affine complete mappings realizing a linear conjugate of* $H_d$ *form exactly* $|\mathcal{R}_d|/n$ *linear-conjugacy classes, each of size* $|\mathrm{GL}(n,2)|$*.*

*Proof.* Corollary 2.4 makes the linear part $A$ of every primitive realization irreducible and its translation nonzero. The field $\mathbb{F}_2[A]$ then identifies $V$ with $\mathbb{F}_{2^n}$; the complement is generated by $a$ and $1+a$ and has some order $d$. The characterization follows from Corollary 2.5. Conversely that corollary realizes every listed $H_d$. Different orders $2^n d$ distinguish the group classes.

A linear transformation normalizes $H_d$ exactly when it normalizes its stabilizer $C_d$ of zero. Since $\mathbb{F}_2[C_d] = \mathbb{F}_{2^n}$, conjugation by such a transformation induces an automorphism of this field, hence a power of Frobenius. After removing that power, the transformation centralizes all scalar multiplications and is itself scalar. Conversely every map $x \mapsto bx^{2^i}$ normalizes $C_d$. The normalizer therefore has the asserted order.

For maps realizing the specified $H_d$, normalizer conjugation sends $(a,c)$ to $(a^{2^i}, bc^{2^i})$. Scalar transformations act transitively on $c \neq 0$. Every $a \in \mathcal{R}_d$ has degree $n$ over $\mathbb{F}_2$: a proper subfield containing $a$ would contain $1+a$ and all of $C_d$. Thus its Frobenius orbit has length $n$, giving $|\mathcal{R}_d|/n$ normalizer orbits. Conjugacy between two maps with group $H_d$ must normalize $H_d$, so these are also the full linear-conjugacy classes when group conjugates are included. Finally, a linear transformation centralizing $f(x) = ax + c$ centralizes the irreducible multiplier $a$, so is scalar; fixing $c \neq 0$ forces it to be the identity. All full linear-conjugacy classes consequently have size $|\mathrm{GL}(n,2)|$. □

For an explicit polynomial encoding, set

$$G_e(t) = \gcd(t^e + 1, (t+1)^e + 1), \qquad P_d(t) = \prod_{e \mid d} G_e(t)^{\mu(d/e)}.$$

Although written as a quotient, $P_d$ is a monic squarefree polynomial over $\mathbb{F}_2$ with root set $\mathcal{R}_d$. Indeed the exponent of a root whose pair generates $C_h$ is $\sum_{h \mid e \mid d} \mu(d/e)$, which is one if $h = d$ and zero otherwise. When $\mathrm{ord}_d(2) = n$, all irreducible factors of $P_d$ have degree $n$ and encode the classes in Proposition S5.2. For example the index-three counts in dimensions $8, 10, 12, 14$ give $3, 9, 33, 123$ map classes, respectively.

The graph interpretation clarifies the distinction between subgroup membership and joint generation. Write

$$\Gamma(q,d) = \mathrm{Cay}(\mathbb{F}_q^+, C_d), \qquad x \sim y \iff y - x \in C_d.$$

This is the usual generalized Paley graph (including the complete graph when $d = q-1$). Its connectivity and ordinary triangle counts are classical [4, 5]. For a triangle with ordered vertices $(x,y,z)$, define its ratio subgroup to be

$$K(\{x,y,z\}) = \langle a, 1+a \rangle, \qquad a = \frac{z-x}{y-x}.$$

This subgroup is independent of the vertex ordering. We call the triangle *generating* when its ratio subgroup is $C_d$. The definition uses normalized ratios, not the subgroup generated by the unnormalized edge differences.

**Proposition S5.3** (Generating triangles and companion realization)**.** *For $d > 1$ dividing $q - 1$, $\Gamma(q, d)$ is connected exactly when $\mathrm{ord}_d(2) = n$. Each edge belongs to $T_d$ ordinary triangles and $|\mathcal{R}_d|$ generating triangles. The total number of generating triangles is $qd|\mathcal{R}_d|/6$. Thus $H_d$ is realizable exactly when the graph is connected and has a generating triangle. If $d > 3$, the action of $H_d$ on generating triangles is free, with $|\mathcal{R}_d|/6$ orbits; in particular $6 \mid |\mathcal{R}_d|$.*

*Proof.* The additive span of $C_d$ is closed under products, contains one, and is a finite subfield. It is therefore $\mathbb{F}_2[C_d] = \mathbb{F}_{2^{\mathrm{ord}_d(2)}}$. The components are its additive cosets, proving the connectivity assertion (compare [4, Theorem 2.2]).

For the edge $\{0, 1\}$, its third vertices are precisely the $a$ with $a, 1 + a \in C_d$, and the generating ones are precisely $a \in \mathcal{R}_d$. Reordering vertices replaces $a$ by one of

$$a, \quad 1 + a, \quad a^{-1}, \quad (1 + a)^{-1}, \quad \frac{a}{1 + a}, \quad \frac{1 + a}{a};$$

these all generate the same subgroup together with one plus themselves. Translations and multiplication by $C_d$ preserve the ratios and act transitively on edges. Counting the $qd/2$ edges and three edges per triangle gives the total. The realization equivalence now follows from Corollary 2.5.

For $d > 3$ the six transforms of $a \in \mathcal{R}_d$ are distinct. A transposition can fix only an excluded value 0 or 1, whereas a three-cycle can fix only $a^2 + a + 1 = 0$, whose pair generates $C_3$. Any affine stabilizer of a generating triangle induces a vertex permutation and would make the corresponding transform fix $a$. It must therefore fix the vertices individually, forcing its multiplier to be one and its translation to be zero. The action is free. Each orbit has size $qd$, giving the orbit count and divisibility. No claim about the full graph automorphism group is required here. □

In the family of Corollary 6.7, the graph is connected and every edge has exactly $t - 2$ third vertices. For an edge $\{x, y\}$ they are $x + (y - x)a$ with $a \in \mathbb{F}_t \setminus \{0, 1\}$. Thus all its triangles lie in the clique $x + (y - x)\mathbb{F}_t$, and none is generating. These graphs are the already studied $(t + 1)$-power graphs $\Gamma_{t,4}(1)$ of Podestá and Videla [6]; their ordinary intersection parameter $t - 2$ follows also from [6, Theorem 5.2]. The descent proof supplies the exact subfield location and its companion-group obstruction. Connectedness and ordinary triangles alone do not guarantee realization. The graph family, character-sum method, and ordinary triangle counts are not claimed as new, and priority of the companion-specific applications remains to be established.

# S6 Single full-order slopes and fiber symmetry

In this section use the quartic notation of main Section 3: $t = 2^s$, $A = t + 1$, $B = t^2 + 1$, $d = AB$, $K = \mathbb{F}_t \subset E = \mathbb{F}_{t^2} \subset L = \mathbb{F}_{t^4}$, $U = \ker N_{E/K}$ and $V = \ker N_{L/E}$. The main paper develops the mixed-character estimate directly. Here we retain the separate single-character argument, the four-torsion fiber symmetry and the single full-order existence proof. These give a second route to the weaker one-slope conclusion.

## S6.1 A single-character estimate

**Lemma S6.1** (Uniform character bound on a norm fiber)**.** *Let $b$ be as in Lemma 3.2, and let $u \in U$ be the unique square root of $b$. For every nontrivial multiplicative character $\chi$ of $V$, one has*

$$\left| \sum_{a \in X_b(K)} \chi(a/u) \right| \le 8\sqrt{t}. \tag{S6.1}$$

*Proof.* We use the classical Weil bound for a geometrically nontrivial finite-order rank-one character on a curve: on a genus-$g$ curve, tame ramification at $r$ geometric points gives the bound $(2g-2+r)\sqrt{t}$, when the character is nontrivial at every one of those points. This is the curve Artin-character estimate discussed in [8, §4 of the author's version]; the Lang-torsor interpretation of finite-field multiplicative characters is also used in [7]. For completeness, the dimension underlying this estimate is $2g-2+r$: compactly supported cohomology in degrees zero and two vanishes, and the tame Euler-characteristic formula gives this dimension in degree one. The trace formula and the Weil theorem for curves bound each eigenvalue by $\sqrt{t}$. For finite-order characters this is a character component of a finite cover of the curve. We now check all these hypotheses here.

Let $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$ denote the two-dimensional norm torus. It has $\mathcal{T}(K) = V$, and $a \mapsto a/u$ maps $X_b$ to $\mathcal{T}$, since $N_{L/E}(u) = u^2 = b$. Over $\overline{K}$ use coordinates $(z_0, z_1)$ on the torus; its Frobenius is

$$F_{\mathcal{T}}(z_0, z_1) = (z_1^{-t}, z_0^t).$$

The Lang isogeny $z \mapsto F_{\mathcal{T}}(z)/z$ has constant kernel $V$. Pulling it back along $a \mapsto a/u$, and writing $r_0 = x/u$, $r_1 = y/u^t$, gives

$$\frac{z_1^{-t}}{z_0} = r_0, \qquad \frac{z_0^t}{z_1} = r_1, \qquad \text{hence} \qquad z_0^B = \frac{r_1^t}{r_0}.$$

Thus this pullback is geometrically a Kummer cover with defining function $h = (y/u^t)^t/(x/u) = y^t/x$, since $u^{t^2} = u$. For a character of exact order $m \mid B$, its quotient cover has equation $z^m = h$ over $\overline{K}$. This geometric Kummer presentation must be distinguished from the $K$-structure of the quotient cover: the latter is inherited from the pullback of the $K$-defined Lang isogeny, with its corresponding Frobenius descent datum. In particular, $h$ need not belong to the function field $K(C_b)$ in these split coordinates. All statements about geometric ramification and geometric connectedness are checked after base change to $\overline{K}$, whereas the character sum is taken using the descended $K$-cover.

The Frobenius descent can be made explicit, so that the character trace is not merely inferred from a convention. On geometric points, the $K$-Frobenius on the base and on the pullback Lang cover is

$$\begin{aligned} F_{C_b}(x, y) &= \left(\frac{b}{y^t}, x^t\right), \\ F_Y(x, y, z_0) &= \left(\frac{b}{y^t}, x^t, \frac{x}{u} z_0\right). \end{aligned} \tag{S6.2}$$

Indeed, the two Lang equations give $z_1^{-t} = r_0 z_0$, while $F_{\mathcal{T}}$ acts on the first split coordinate as $z_0 \mapsto z_1^{-t}$. The displayed formula is also compatible with the Kummer equation, since $u^B = b$ and

$$h\left(\frac{b}{y^t}, x^t\right) = \frac{x^{t^2} y^t}{b} = \left(\frac{x}{u}\right)^B h(x, y).$$

For a $K$-rational point, $(x, y) = (a, a^t)$ with $a^B = b$; thus Frobenius fixes the base point and acts on its geometric Lang fiber by $z_0 \mapsto (a/u) z_0$. This is precisely the deck transformation indexed by $a/u \in V$. In the character eigensheaf associated with $\chi$, its Frobenius trace is therefore $\chi(a/u)$, or $\chi^{-1}(a/u)$ if the inverse Frobenius convention is used. The latter choice is immaterial for the asserted bound. In particular, this computation verifies the Frobenius trace identification for the actual $K$-descent of the cover, rather than for the bare geometric Kummer equation.

The zeros and poles of $x$ are simple and supported on $D_x$; those of $y$ are simple and supported

on the disjoint set $D_y$. The valuations can be read point by point:

| points | $\mathrm{ord}_P(h)$ |
|---|---|
| $(0,1),\ (0,b^{-1})$ | $-1$ |
| $(\infty,1),\ (\infty,b^{-1})$ | $1$ |
| $(1,0),\ (b,0)$ | $t$ |
| $(1,\infty),\ (b,\infty)$ | $-t$ |

Thus the eight nonzero valuations are $1,-1,t,-t$, each twice. Since $m$ is odd and $\gcd(m,t)=1$, the character is tamely and nontrivially ramified at all eight points. A valuation of absolute value one also makes this cover geometrically connected, excluding a constant-field character or a geometrically trivial pullback. The conductor degree is consequently exactly eight, independently of $m,t,b$. Lemma 3.2 and the curve bound now give $(2\cdot 1-2+8)\sqrt{t}=8\sqrt{t}$. □

## S6.2 The four-torsion symmetry

**Proposition S6.2** (Four-torsion symmetry of a norm fiber)**.** *In the setting of Lemma 3.2, let $u\in U$ satisfy $u^2=b$. On the geometric $(x,y)$-model of $C_b$ set*

$$\rho(x,y)=\left(\frac{1}{y},\frac{x}{b}\right),\qquad h=\frac{(y/u^t)^t}{x/u}=\frac{y^t}{x}.$$

*Then $\rho$ descends to a $K$-automorphism of order four. After choosing a $K$-rational origin, it is translation by a $K$-rational point of exact order four. The eight boundary points form exactly two $\rho$-orbits of length four:*

$$(0,1)\longmapsto(1,0)\longmapsto(\infty,b^{-1})\longmapsto(b,\infty)\longmapsto(0,1),$$
$$(0,b^{-1})\longmapsto(b,0)\longmapsto(\infty,1)\longmapsto(1,\infty)\longmapsto(0,b^{-1}).$$

*Moreover, as identities in the geometric function field,*

$$\rho^*h=(uy)^Bh^{-t},\qquad(\rho^2)^*h=h^{-1}. \tag{S6.3}$$

*For every character $\chi$ of $V$, the sums in Lemma S6.1 satisfy*

$$S_b(\chi):=\sum_{a\in X_b(K)}\chi(a/u)=S_b(\chi^{-t})=S_b(\chi^{-1})\in\mathbb{R}.$$

*Every nontrivial character belongs to an orbit of length four under $\chi\mapsto\chi^{-t}$. If $N_b^{\mathrm{prim}}$ denotes the number of $a\in X_b(K)$ with $\mathrm{ord}(a)=d$, then $4\mid\#X_b(K)$ and $4\mid N_b^{\mathrm{prim}}$.*

*Proof.* The field automorphism $\sigma:a\mapsto a^t$ of $L/K$ extends as $\sigma\otimes 1$ to an $R$-linear automorphism of $L\otimes_K R$ for every $K$-algebra $R$; composing this extension with inversion gives a $K$-automorphism of the ambient restriction-of-scalars torus. On $K$-points its action is $a\mapsto(a^t)^{-1}$. Here this notation refers to the Galois automorphism $\sigma$, not the relative Frobenius morphism on the torus. The functorial norm identities

$$N_{L/E}((\sigma a)^{-1})=b^{-t}=b,\qquad N_{L/K}(1+(\sigma a)^{-1})=\frac{N_{L/K}(1+a)}{N_{L/K}(a)}=1$$

show that it preserves both defining conditions, since $N_{L/K}(a)=N_{E/K}(b)=1$. It gives the displayed $\rho$ in split coordinates. Equivalently, the split-coordinate $K$-Frobenius descent is $(x,y)\mapsto(b/y^t,x^t)$, with which $\rho$ commutes. Direct iteration gives

$$\rho^2(x,y)=(b/x,b^{-1}/y),\qquad\rho^4=1.$$

The eight boundary points are paired without fixed points by $\rho^2$. Any remaining geometric fixed point would have $x^2 = b$, $y^2 = b^{-1}$, hence $(x, y) = (u, u^{-1})$. Substitution in the equation of $C_b$ forces $b + b^{-1} = 1$, equivalently $u^2 + u + 1 = 0$, whence $b^3 = 1$, contrary to $b^3 \neq 1$. Thus $\rho^2$, and hence $\rho$, has no geometric fixed point. The bound $\#C_b(K) \geq t + 1 - 2\sqrt{t} > 0$ supplies a rational origin. After choosing a rational origin, any automorphism of a genus-one curve is a translation followed by an origin-preserving automorphism. If the latter is nontrivial, its difference from the identity is a nonzero, hence surjective, elliptic-curve endomorphism, so the composite has a geometric fixed point. Consequently $\rho$ is a translation, necessarily by a $K$-rational point of exact order four.

The two boundary cycles follow by substitution in the rational formula, interpreted on $\mathbb{P}^1 \times \mathbb{P}^1$. Since $u^{t+1} = 1$, $u^B = u^2 = b$ and $b^t = b^{-1}$,

$$\rho^* h = \frac{(x/b)^t}{1/y} = bx^t y = (uy)^B \left(\frac{y^t}{x}\right)^{-t},$$

while the formula for $\rho^2$ yields $(\rho^2)^* h = h^{-1}$. For $a = uv$ with $v \in V$, $\rho$ sends $a$ to $uv^{-t}$. It therefore permutes $X_b(K)$ and preserves $\mathrm{ord}(a)$; changing variables in the character sum proves the stated equalities. Finally $B = t^2 + 1$ is odd, $(-t)^2 \equiv -1 \pmod{B}$. For a character of order $m > 1$, an orbit of length one or two would imply $m \mid 2$, which is impossible since $m \mid B$ is odd. Thus every nontrivial character has an orbit of length four. Translation by a point of exact order four acts freely, proving both divisibilities. □

These symmetries do not by themselves reduce the bound in Lemma S6.1: grouping the characters into four-element orbits leaves the total absolute sieve coefficients unchanged. The estimate used below remains $8\sqrt{t}$.

### S6.3 The single full-order statistic

**Theorem S6.3** (Primitive slopes in the inner critical family)**.** *Put $\theta(B) = \varphi(B)/B$ and $W(B) = 2^{\omega(B)}$. For each $b \in U$ of order $A$, let*

$$N_b^{\mathrm{prim}} = \#\{a \in X_b(K) : \mathrm{ord}(a) = d\}.$$

*Then*

$$\left|N_b^{\mathrm{prim}} - \theta(B)(t+1)\right| \leq \theta(B)(8W(B) - 6)\sqrt{t}. \tag{S6.4}$$

*In particular, with $N_d^{\mathrm{prim}} = \#\{a \in C_d : \mathrm{ord}(a) = d,\ 1 + a \in C_d\}$,*

$$\left|N_d^{\mathrm{prim}} - \varphi(A)\theta(B)(t+1)\right| \leq \varphi(A)\theta(B)(8W(B) - 6)\sqrt{t}. \tag{S6.5}$$

*Thus $N_d^{\mathrm{prim}} \sim \varphi(A)\theta(B)t$ as $s \to \infty$.*

*Proof.* Every $a \in X_b(K)$ has a unique expression $a = uv$ with $v \in V$, where $u$ is the square root of $b$ in $U$. Since $\mathrm{ord}(u) = A$ and $\gcd(A, B) = 1$, it has order $d$ exactly when $v$ has order $B$. The usual character indicator in a cyclic group of order $B$ is

$$\mathbf{1}_{\mathrm{ord}(v)=B} = \theta(B) \sum_{m \mid B} \frac{\mu(m)}{\varphi(m)} \sum_{\mathrm{ord}(\chi)=m} \chi(v).$$

Indeed the sum factors over primes $p \mid B$: the corresponding factor is zero on $V^p$ and $p/(p-1)$ off $V^p$, which cancels the factor $(p-1)/p$ in $\theta(B)$. The trivial-character term is $\theta(B)\#X_b(K)$.

For each nontrivial squarefree divisor $m$ the sum of the absolute values of its coefficients is one. There are $W(B) - 1$ such divisors. Lemmas 3.2 and S6.1 therefore give an error of at most $\theta(B)(2 + 8(W(B) - 1))\sqrt{t}$, proving (S6.4).

An element of order $d$ has relative norm of order $A$, because $N_{L/E}(a) = a^B$. Hence these $\varphi(A)$ norm fibers partition the elements counted by $N_d^{\mathrm{prim}}$. Summing proves (S6.5). Finally $2^{\omega(N)} = N^{o(1)}$: for any $\epsilon > 0$, all but finitely many primes satisfy $2 \le p^\epsilon$, and the remaining primes contribute only a constant. Since $B \asymp t^2$, the relative error tends to zero. □

**Theorem S6.4** (All-dimensional realization at index $t - 1$)**.** *For every $s \ge 2$, there exists $a \in \mathbb{F}_{2^{4s}}$ with*

$$\mathrm{ord}(a) = (2^s + 1)(2^{2s} + 1), \qquad (1 + a)^{(2^s+1)(2^{2s}+1)} = 1.$$

*Consequently $\mathcal{R}_d \neq \varnothing$ at $k = t - 1$ for every such $s$. The proper primitive group $L^+ \rtimes C_d$ is generated by $f(x) = ax + c$ and $F(f)$ for every $c \in L^\times$.*

*Proof.* Every prime divisor $p$ of $B = t^2 + 1$ is 1 modulo 4, since $t^2 \equiv -1 \pmod p$. If $p \ge 256$, then $2 \le p^{1/8}$. For the smaller primes a direct integer product gives

$$\prod_{\substack{p<256 \text{ prime} \\ p\equiv 1 \pmod 4}} \frac{256}{p} < 32^8.$$

The 24 primes in this product and the exact integer comparison are supplied with the finite verifier in Appendix A. It follows for every $B$ under consideration that

$$W(B) < 32B^{1/8}.$$

For $s \ge 33$, $t^2 \ge 2^{66}$ and $t^2 + 1 < 2t^2$ imply $t^4 > 2^{64}(t^2 + 1)$. Equivalently, $\sqrt{t} > 256B^{1/8} > 8W(B)$. The lower bound in (S6.4) is then positive for every primitive $b \in U$.

For $2 \le s \le 32$, Appendix A gives explicit field towers $E = \mathbb{F}_2[X]/(M_s)$ and $L = E[Z]/(Z^2 + cZ + b)$, in which $a = Z$ is a certificate. Irreducibility, both norm conditions and the exact order $d$ are verified by deterministic finite arithmetic, with complete factorizations and recursively certified primes. These 31 cases exhaust the remaining range; they are not an extrapolation.

Finally $d > t^2 - 1$ implies $\mathbb{F}_2[a] = L$: every proper subfield of $L$ has at most $t^2$ elements. Thus $C_d$ acts irreducibly, while $k = t - 1 > 1$ makes it a proper multiplier group. The affine conclusion follows from Corollary 2.5. □

The two adjacent indices $t - 1$ and $t + 1$ therefore have opposite realization behavior for every $s \ge 2$. This does not settle the universal question for all admissible $k \le q^{1/4}$. Also, $N_d^{\mathrm{prim}}$ counts primitive individual slopes; it is a subset count within $|\mathcal{R}_d|$. The joint and double full-order counts in the main paper use the mixed-character estimate. Ordinary intersections for index $t - 1$ belong to cyclotomic number theory, including the general norm-based treatment of Kudo and Nogata [16]. The additional exact-order condition is the focus here. The classical Weil bound, Lang construction, and primitive-character sieve are attributed above; priority for this particular realization theorem is not asserted.

## S7 Prescribed-order predecessors and quantitative comparison

Put $t = 2^s$, $s \ge 2$, $A = t + 1$, $B = t^2 + 1$, $d = AB$ and $r = t - 1$. The identities $d = (t^4 - 1)/r$ and $C_d = \ker N_{L/K}$ identify the double full-order statistic with an established $r$-primitive-pair count. This section compares particular published estimates at these parameters. It does not assert that failure of a sufficient criterion excludes every refinement of its proof method.

### S7.1 Adjacent exact-order pairs

In [14, Proposition 3.1], set the field size to $t^4$, the index to $r$, the common difference to 1 and the length to 2. Its $N_r(t^4, 1, 2)$ equals $D_d$. Write $M = \theta(d)^2(t+1)^2$. Their estimate is centered at

$$M_{\mathrm{SSL}} = \theta(d)^2 \frac{t^4}{r^2}, \qquad M_{\mathrm{SSL}} - M = \theta(d)^2 \frac{2t^2 - 1}{r^2} = O(1).$$

Thus the leading density is already represented by that estimate. Its relative error bound becomes $((rW(d))^3/(rW(d)-1))/t^2$. Theorem 1 there would require

$$t^2 \geq \frac{(rW(d))^3}{rW(d) - 1}.$$

This fails: $\gcd(A, B) = 1$, $A, B > 1$, so $W(d) \geq 4$, and the right-hand side exceeds $16r^2 > t^2$. Its stronger Theorem 5 sieve cannot repair this numerical loss. Even allowing length two in that displayed criterion, let $h$ be the number of omitted primes, $l \mid d$ the sieve core, $\delta = 1 - 2\sum_{i=1}^{h} p_i^{-1} > 0$ and $\Delta = (2h-1)/\delta + 2$. It requires

$$t^2 \geq \Delta r^2 W(l)^2.$$

If $h = 0$, then $\Delta = 1$ and $W(l) = W(d) \geq 4$. If $h \geq 1$, then $\Delta \geq 3$ and $W(l) \geq 1$. In either case its right-hand side exceeds $t^2$ for $t \geq 4$. This checks every legal split of that criterion, independently of the length restriction preceding the sieve section. The fixed-$r$ asymptotic corollary does not by itself give a uniform conclusion for $r = t - 1$.

For comparison, [10, Corollary 17], with polynomials $X, X+1$, indices $r, r$ and freeness parameters $d, d$, requires $t^2 > r^2 W(d)^2$, which also fails. The general $r$-primitive and $k$-normal frameworks in [11, Theorem 3.2] and [12, Theorem 3.3] have additional conditions; even their favorable numerical specialization with trivial additive factors has right-hand side $2r^2W(d)^2 > t^2$. These comparisons concern the stated sufficient inequalities, not a new formulation of adjacent prescribed orders.

### S7.2 Prescribed norms and a change of base field

Choudhary–Sharma [15, Definition 3.1] give another exact identification. Set their $q = t$, $n = 4$, $m = 2$, $\gamma = 1$, $r_1 = r_2 = r$, and $a_1 = a_2 = 1$. Then $r_i' = r$, $R_i = d$, $R_i' = 1$, $Q_{R_i} = d$, and $N_{1,1}(d) = D_d$. Their Theorem 1 becomes $1 > 2r^2W(d)^2$. For Theorem 3 write $U = \sum_i u_i$ and $\Delta = (U-1)/\delta + 2$. If $U = 0$ then $\Delta = 1$; otherwise $\Delta \geq 2$. Its right-hand side is at least $2r^2$. For every legal split in Theorem 4, the positive denominator is at most 1 and the numerator is at least $\sum_i c_i \geq 2$; the full right-hand side is at least $4r^2$.

Changing the base field cannot rescue these inequalities. If $q_0^n = t^4$, their left-hand side for length two is

$$q_0^{n/2-2} = \frac{t^2}{q_0^2} \leq \frac{t^2}{4} < 2r^2.$$

This already lies below the lower bounds for Theorems 1, 3 and 4, even when all required norm data are admissible. Their Theorem 2 assumes odd characteristic. The application with $r = 3$ is irrelevant for $s \geq 3$. When $s = 2$, the only possible base with $n \geq 7$ is $(q_0, n) = (2, 8)$, which is listed among its possible exceptions in Theorem 5.

## S7.3 The reciprocal-trace statistic

The count $\Delta_t$ in Proposition S8.1 is exactly $N_{t^4}(B, A)$ in [18, Theorem 1.2]: order $B$ forces $\gamma \in V$, and order $A$ forces $\gamma + \gamma^{-1} \in U$. Substitution in that theorem gives

$$\left|\Delta_t - \frac{\theta(d)}{t-1}\right| < 2\theta(d)W(d)t^2.$$

The center is much smaller than the actual norm-concentrated population, but the large error permits this. The estimate here is

$$|\Delta_t - \theta(d)(t+1)| \le \theta(d)(4W(d) - 2)\sqrt{t}.$$

It is the refined geometry of the same statistic that changes the estimate; the statistic itself is prior. The substitution $a = (1+\gamma)^{-1}$ and its Frobenius relation also occur in [18, §4]. Theorem 1.5 there, with base $t^2$, prescribes reciprocal-trace order $t^2 - 1$, not $A$; that trace has norm of order $t - 1$ to $K$.

## S7.4 Quartic geometry and the exact affine slice

Hachenberger's formal quartic treatment [13] gives a direct geometric predecessor. In his notation set $q = t$, with his ground, intermediate and quartic fields corresponding to our $K, E, L$, respectively. Section 3.1 (p. 94) gives $L^\times = K^\times \times C_d$ and identifies primitive projective points with order-$d$ norm-one representatives. Proposition 4.1 and Remark 4.2 (p. 97) give the ovoids

$$\mathcal{O}_u = \{[uv] : v \in V\}, \qquad u \in U.$$

They are primitive exactly when $\mathrm{ord}(u) = A$; each then contains $\varphi(B)$ primitive points in all. Proposition 7.1 (p. 107) shows that every $u \neq 1$ has a trace-zero oval $\mathcal{O}_u \cap H$ with $t+1$ points, where $H = \mathbb{P}(\ker \mathrm{Tr}_{L/K})$. These statements exactly overlap the ambient geometry used here.

The remaining condition has a precise projective formulation. For a primitive point $[x] \in H$, put $c = x + x^{t^2}$. Then $c \in K^\times$: trace transitivity puts $c$ in $K$, and $c = 0$ would put $x$ in the quadratic subfield, contrary to primitivity. Define

$$\rho([x]) = \frac{N_{L/K}(x)}{(x + x^{t^2})^4} \in K^\times. \tag{S7.1}$$

Replacing $x$ by $\lambda x$, $\lambda \in K^\times$, multiplies numerator and denominator by $\lambda^4$; thus $\rho$ is well defined. The trace-one representative $a = x/c$ has $N_{L/K}(a) = \rho([x])$. Consequently the trace-one double full-order subfamily is exactly the primitive-point level set $\rho = 1$. Indeed $\rho = 1$ gives $a \in C_d$ of order $d$ and $a^{t^2} = 1 + a$. Conversely, such an $a$ gives $\gamma = (1+a)/a$ of order $B$ and $\gamma + \gamma^{-1} = a^{-B}$ of order $A$, recovering Proposition S8.1.

This also locates the global primitive-normal bounds relative to the count needed here. Let $M_c$ count order-$d$ norm-one representatives with $a + a^{t^2} = c$, $c \in K^\times$. Characteristic two makes the trace-zero plane precisely the non-normal configuration [13, §5.2.1]. Each primitive projective point has $\varphi(t-1)$ ambient primitive representatives. With $PN_4(t)$ as in [13], therefore,

$$\sum_{c \in K^\times} M_c = \varphi(d) - \frac{PN_4(t)}{\varphi(t-1)}, \qquad M_1 = \Delta_t. \tag{S7.2}$$

Section 6.1 and Theorem 1.2 there give $\sum_c M_c \le t\varphi(A)$. Theorem 1.3 makes this an equality when $B$ is prime. Neither is a positive lower bound on $M_1$. For a norm-one representative, $\rho = c^{-4}$, so division by $c$ preserves norm one exactly when $c = 1$.

The distinction is visible in exact small parameters. For $t = 8$ and 32, the primitive trace-zero populations in (S7.2) are 36 and 500, respectively, while $M_1 = \Delta_t = 0$ in both cases. The full double-order populations are nevertheless $D_d = 24$ and 240, respectively. Thus the nonempty primitive-ovoid oval cannot by itself prove positivity of the fixed trace-one subfamily. The estimates of Proposition S8.1 and Theorems 4.3–4.4 are not stated in that projective treatment; the additional affine distribution and order filters require further analysis. This comparison checks its full formal text and the relevant proofs; it does not claim an exhaustive survey of all subsequent refinements of quartic geometry.

Kudo–Nogata [16, Propositions 3.6 and 4.3] treat quadratic norm root counts and the ordinary cyclotomic intersection, respectively. Setting their base to $t$ and extension degree to 4 gives the unfiltered membership count $T_d$. The extra joint-generation and individual-order filters require further control. In particular, Proposition 3.1 is the elementary quadratic root-count reduction, rather than the main quantitative innovation.

The curve character bound, Lang construction, and primitive indicators used here are classical [8, 7, 10]. The proofs establish their hypotheses on the specified fibers: the constant norm-character exceptions, geometric monodromy, active boundary conductors and the exact coefficient average. Together with the all-parameter closure, these are the results on which the present contribution is based.

### S7.5 Trace statistics in norm tori

Theorem 5.1 extends the trace-one calculation to every $c \in E^*$, with the uniform four-point bound. Theorem 5.4 classifies the complete empty groups $C_9$ at $s = 3$ and $C_{11}$ at $s = 5$. The current finite proof uses the inverse-support reduction, 755 representative witnesses and 731 separating witnesses; the older all-trace coverage families remain a separate check. Proposition 5.5 and Corollaries 5.6–5.7 give the within-class distribution and arbitrary-norm transfer. Main Appendix B states the current certificate dependencies.

There is related recent work on trace fibers of norm tori. Shin [29, Proposition 5.6 and Theorems 5.7–5.8] treats finite étale cubic algebras in characteristic different from $2, 3$. Every smooth trace/norm fiber is equidistributed over subgroup cosets, with error at most $3(1 - 1/m)\sqrt{q}$ for index $m$. The proof uses Lang character sheaves and a Picard–Kummer kernel calculation; the nodal boundary has exceptional characters. Thus square-root trace equidistribution in norm tori and its geometric character analysis are already part of the literature. Those stated cubic theorems do not apply to the present characteristic-two quartic relative-trace curves. The specific descent, four boundary valuations, and exact-order classification are the results established here, rather than a new general sheaf method.

Rani–Sharma–Tiwari–Panigrahi [30] provide a prescribed-order, norm and trace framework containing these counts after specialization and summation. The relevant sufficient conditions in their Section 4 become $t^{-1} > (t-1)W(d)$ over $K$, or $t^{-2} > (t-1)W(B)$ over $E$, and fail for $t \geq 4$. For $c \in K^*$ the relative-trace population is 1-normal over $K$; for $c \in E \setminus K$ it is normal. A condition on absolute trace alone therefore does not prescribe the individual $E$-valued relative trace.

The earlier function-field count [31] and quartic primitive-normal geometry [32] are also predecessors. Main Appendix D gives the exact parameter comparisons, including genus refinements and the aggregate-to-slice identity, and compares the product, hypersurface and rational-image criteria. These comparisons concern the stated results at the specified parameters; failure of a displayed sufficient criterion does not exclude further refinements of its methods or establish absolute priority.

## S8 The trace-one curve and its Lang covers

Let $t = 2^s$, $s \geq 2$, $K = \mathbb{F}_t$, $E = \mathbb{F}_{t^2}$, $L = \mathbb{F}_{t^4}$, $A = t + 1$, $B = t^2 + 1$ and $d = AB$. Write $U = \ker N_{E/K}$ and $V = \ker N_{L/E}$. The trace-one calculation is the $c = 1$ instance of Theorem 5.1. It is useful also as a direct analysis of the parameter excluded by the good-fiber hypothesis $b^3 \neq 1$ in Proposition 3.4. The next proof specifies the Lang multipliers and all four inertia characters at this parameter. The stronger positivity range and the complete list of empty trace-one slices are given by Theorem 5.4.

**Proposition S8.1** (A trace-one double full-order subfamily)**.** *For $t = 2^s$, $s \geq 2$, use $A = t + 1$, $B = t^2 + 1$ and $d = AB$ as above. Put*

$$\Delta_t = \#\{\gamma \in V : \tau = \gamma + \gamma^{-1} \in U,\ \mathrm{ord}(\gamma) = B,\ \mathrm{ord}(\tau) = A\}.$$

*Every element counted by $\Delta_t$ gives, injectively,*

$$a = (1 + \gamma)^{-1} \in C_d, \qquad 1 + a = a^{t^2}, \qquad \mathrm{ord}(a) = \mathrm{ord}(1 + a) = d. \tag{S8.1}$$

*In particular $D_d \geq \Delta_t$. For any characters $\eta$ of $U$ and $\chi$ of $V$ with $(\eta, \chi) \neq (1, 1)$,*

$$\left| \sum_{\substack{\gamma \in V \\ \gamma + \gamma^{-1} \in U}} \eta(\gamma + \gamma^{-1}) \chi(\gamma) \right| \leq 4\sqrt{t}. \tag{S8.2}$$

*Consequently*

$$|\Delta_t - \theta(d)(t + 1)| \leq \theta(d)(4W(d) - 2)\sqrt{t}. \tag{S8.3}$$

*The sufficient condition $\sqrt{t} \geq 4W(d)$ implies $\Delta_t > 0$; it holds for every $s \geq 33$ by the integer bounds in Appendix A.*

*Proof. The full-order construction.*

Here $\gamma^{t^2} = \gamma^{-1}$ and $\gamma \neq 1$. Thus $a^{t^2} = \gamma/(1 + \gamma) = 1 + a$. Also

$$N_{L/E}(1 + \gamma) = \gamma + \gamma^{-1} = \tau, \qquad a^B = \tau^{-1}, \qquad a^{t^2 - 1} = \gamma.$$

Since $N_{E/K}(\tau) = 1$, the norm of $a$ to $K$ is one, so $\mathrm{ord}(a)$ divides $d$. The two displayed powers force both $A$ and $B$ to divide this order. Their gcd is one, and hence the order is $d$. The conjugate $1 + a$ has the same order. The formula for $a$ is injective in $\gamma$.

*The curve and its four boundary points.*

We analyze the norm fiber $X_1$ directly, rather than applying Proposition 3.4, whose good-fiber hypothesis excludes this parameter. Write its variable as $\gamma$. Over $\overline{K}$, the relative norm-one torus has coordinates $(x, y, x^{-1}, y^{-1})$. Its $K$-points have $x = \gamma$, $y = \gamma^t$, $\gamma \in V$, and its Frobenius is

$$F_Y(x, y) = (y^{-t}, x^t).$$

The condition $N_{E/K}(\gamma + \gamma^{-1}) = 1$ gives the projective closure $Y_{\overline{K}} \subset \mathbb{P}^1 \times \mathbb{P}^1$ of

$$(x + 1)^2 (y + 1)^2 = xy. \tag{S8.4}$$

Here $Y$ denotes the $K$-curve with the inherited torus descent $F_Y$, not the split curve equipped with coordinatewise $t$-power Frobenius. The descent extends to the projective closure and preserves its

boundary. In particular the relation $y = x^t$ is imposed on $K$-points, not in its geometric function field. On the affine chart the two partial derivatives are $y$ and $x$; their only common zero $(0, 0)$ is not on the curve. The boundary is

$$P_0 = (0, 1), \quad P_\infty = (\infty, 1), \quad Q_0 = (1, 0), \quad Q_\infty = (1, \infty).$$

At $P_0$, with $u = y+1$, the equation is $(1+x)^2u^2+x(1+u) = 0$, whose derivative with respect to $x$ is one at the point. Thus it is smooth and $v_{P_0}(x) = 2$. Reciprocating $x$ or $y$, and interchanging them, gives the same smoothness test at the other boundary points. This proves that the bidegree-$(2, 2)$ curve is smooth. It is geometrically irreducible. Smoothness implies that the curve is reduced and that distinct irreducible components would be disjoint. Every component has a nonzero effective bidegree $(a_i, b_i)$, with $a_i, b_i \geq 0$ and $\sum_i(a_i, b_i) = (2, 2)$. If one component has both entries positive, its intersection $a_ib_j + a_jb_i$ with every other component is positive. If no component has both entries positive, the sum $(2, 2)$ forces a pure vertical component and a pure horizontal component, whose intersection is again positive. Both possibilities contradict disjointness. Adjunction on $\mathbb{P}^1 \times \mathbb{P}^1$ now gives genus $(2 - 1)(2 - 1) = 1$. Frobenius permutes the four boundary points in the cycle

$$P_0 \longmapsto Q_0 \longmapsto P_\infty \longmapsto Q_\infty \longmapsto P_0.$$

There is no $K$-rational boundary point. Therefore the open curve $Y^\circ$ has exactly the points in the sum (S8.2), and $|\#Y^\circ(K) - (t + 1)| \leq 2\sqrt{t}$ by Hasse–Weil.

*Actual Lang covers and their character traces.*

Let $\mathcal{U} = \ker N_{E/K}$ and $\mathcal{V} = \mathrm{Res}_{E/K}(\ker N_{L/E})$; their $K$-point groups are $U$ and $V$. On $Y^\circ$ use the maps $\tau = \gamma + \gamma^{-1}$ to $\mathcal{U}$ and $\gamma$ to $\mathcal{V}$ itself. Write

$$w_0 = (x + 1)^2/x, \qquad h = y^t/x.$$

The split coordinates of $\tau$ are $(w_0, w_0^{-1})$. The two Lang pullbacks, with their inherited $K$-structures, are

$$z_U^A = w_0^{-1}, \qquad z_V^B = h. \tag{S8.5}$$

For the first, the Lang equation is $z_U^{-t}/z_U = w_0$. For the second, the split equations are $z_1^{-t}/z_0 = x$, $z_0^t/z_1 = y$; elimination gives $z_0^B = y^t/x$, with $z_V = z_0$. Thus the actual functions are $w_0^{-1}$ and $h$, not merely a geometrically convenient Kummer model. Above $F_Y$, Frobenius on the two covering coordinates is

$$(z_U, z_V) \longmapsto (w_0z_U, xz_V).$$

This follows from the split Lang equations. It also respects (S8.5), since $w_0(F_Y) = w_0^{-t}$ and $h(F_Y) = x^Bh$. At a $K$-point the deck multipliers are exactly $\tau$ and $\gamma$. Choosing the character convention for these multipliers gives trace $\eta(\tau)\chi(\gamma)$; the inverse convention just inverts both characters.

*Geometric connectedness and conductor.*

The complete boundary valuation table is

| | $w_0$ | $h$ |
|---|---|---|
| $P_0$ | $-2$ | $-2$ |
| $P_\infty$ | $-2$ | $2$ |
| $Q_0$ | $2$ | $2t$ |
| $Q_\infty$ | $2$ | $-2t$ |

(S8.6)

For example $x+1$ has simple zeros at $Q_0, Q_\infty$ and a double pole at $P_\infty$; $y$ has a double zero at $Q_0$ and a double pole at $Q_\infty$. These observations give every entry of the table. Suppose for $0 \le e_0 < A$, $0 \le e_1 < B$ that

$$(w_0^{-1})^{Be_0} h^{Ae_1} \in \overline{K}(Y)^{\times AB}.$$

Its valuation at $P_0$ is $2Be_0 - 2Ae_1$. Reduction modulo $A$ and modulo $B$, using that $A, B$ are odd and coprime, gives $A \mid e_0$ and $B \mid e_1$. This single valuation separates the two cyclic components because reduction modulo each order kills the other component. Thus $e_0 = e_1 = 0$. The Kummer criterion proves that the product cover in (S8.5) is geometrically connected with full group $U \times V$; in particular every nontrivial character quotient is geometrically nontrivial.

All cover degrees are odd, so the ramification is tame. At every boundary point the valuation of $w_0^{-1}$ is one of $\pm 2$, and that of $h$ is one of $\pm 2$ or $\pm 2t$. Since $A$ and $B$ are odd, $\gcd(2, A) = \gcd(2t, B) = 1$; each coordinate therefore generates its cyclic deck factor. A nontrivial character on either factor is nontrivial on that coordinate, and the two character values cannot cancel because their orders divide the coprime integers $A$ and $B$. Thus every nontrivial pair is ramified at each of the four boundary points, with Swan conductor zero and local Artin conductor one. Its total conductor is exactly four, and it is unramified on $Y^\circ$. The associated rank-one finite-monodromy sheaf is pure of weight zero and geometrically nontrivial. Thus $H_c^0 = H_c^2 = 0$, and Grothendieck–Ogg–Shafarevich gives $\dim H_c^1 = 2g(Y) - 2 + 4 = 4$. The classical curve character bound [8, §4, Lemma 1 in the author's version], with the Lang construction as in [7], bounds each Frobenius eigenvalue by $\sqrt{t}$. This proves (S8.2).

*The two primitive indicators and positivity.*

Apply the primitive-character indicator to $\tau \in U$ and $\gamma \in V$. Their product has constant coefficient $\theta(A)\theta(B) = \theta(d)$ and total absolute coefficient sum $\theta(d)W(A)W(B) = \theta(d)W(d)$. The trivial pair has Hasse error $2\theta(d)\sqrt{t}$. The other pairs have total coefficient mass $\theta(d)(W(d) - 1)$ and character bound $4\sqrt{t}$. Their sum is exactly the error in (S8.3). If $\sqrt{t} \ge 4W(d)$, then $(t+1) - (4W(d) - 2)\sqrt{t} > 0$, proving positivity. The independent integer bound in main Lemma A.1 supplies this condition for every $s \ge 33$, proving $\Delta_t > 0$ throughout that range. □

The reciprocal-trace substitution is classical [18, §4]. The argument above verifies its descended norm geometry and its four-point conductor; it does not assign priority to the substitution or to the general curve bound.

## S.A Self-contained proof for the even-characteristic case

*Proof of Theorem S2.2.* The statement is classical (see [19] and [20, Theorem 1, p. 1]; it also follows from the equivalent square-order formulation in [18, Theorem 1.5, p. 170]); we give a self-contained proof in the even-characteristic case used above. Cohen and Presern [21] also give a self-contained treatment of prescribed traces. This appendix reconstructs the classical trace-one argument and makes its finite verification explicit; no priority is claimed for the existence result or the character-sum method. Put $q = 2^k$, $L = \mathbb{F}_{q^2}$, $K_0 = \mathbb{F}_q$, and $M = q^2 - 1 = (q-1)(q+1)$. Let

$$E_1 = \{x \in L : x^q + x = 1\}.$$

The trace map $L \to K_0$ is surjective and $K_0$-linear, so $|E_1| = q$. If $x \in E_1$, then $x \notin K_0$ and its minimal polynomial over $K_0$ is $X^2 + X + xx^q$. Hence it is enough to prove that $E_1$ contains an element of order $M$.

Let $\psi$ be the nontrivial additive character of $\mathbb{F}_2$, and put $\psi_L(x) = \psi(\mathrm{Tr}_{L/\mathbb{F}_2}(x))$ and $\psi_{K_0}(a) = \psi(\mathrm{Tr}_{K_0/\mathbb{F}_2}(a))$. For a multiplicative character $\chi$ of $L^*$, extend $\chi$ to 0 by $\chi(0) = 0$ and define

$$g_L(\chi) = \sum_{x \in L^*} \chi(x)\psi_L(x), \qquad g_{K_0}(\chi) = \sum_{a \in K_0^*} \chi(a)\psi_{K_0}(a), \qquad S(\chi) = \sum_{x \in E_1} \chi(x).$$

If $\chi$ is nontrivial on $L^*$, then $|g_L(\chi)| = q$; if $\chi|_{K_0^*}$ is nontrivial, then $|g_{K_0}(\chi)| = \sqrt{q}$. Indeed, the usual squared-modulus calculation gives

$$|g_L(\chi)|^2 = \sum_{t \in L^*} \chi(t)^{-1} \sum_{a \in L^*} \psi_L(a(1-t)) = q^2,$$

and the same argument in $K_0$ gives $q$.

We first compute $S(\chi)$. For $\alpha \in K_0$ set $E_\alpha = \{x \in L : x^q + x = \alpha\}$; these sets partition $L$, and $E_\alpha = \alpha E_1$ for $\alpha \neq 0$. If $\chi^{q+1} \neq 1$, then the restriction of $\chi$ to $K_0^*$ is nontrivial. Since $E_0 = K_0$, its contribution is $\sum_{x \in E_0} \chi(x) = 0$; hence

$$g_L(\chi) = \sum_{\alpha \in K_0} \psi_{K_0}(\alpha) \sum_{x \in E_\alpha} \chi(x) = S(\chi) g_{K_0}(\chi),$$

so $|S(\chi)| = q/\sqrt{q} = \sqrt{q}$.

Suppose now that $\chi^{q+1} = 1$ and $\chi \neq 1$. Then $\chi|_{K_0^*} = 1$. Let $H = \{h \in L^* : h^{q+1} = 1\}$; since $\gcd(q-1, q+1) = 1$, the group $L^*$ is the direct product $K_0^* \times H$. Write $x = ch$ with $c \in K_0^*$ and $h \in H$. The condition $x^q + x = 1$ becomes $c(h + h^{-1}) = 1$; thus $h \neq 1$ and $c = (h + h^{-1})^{-1}$. This gives a bijection between $H \setminus \{1\}$ and $E_1$, and therefore

$$S(\chi) = \sum_{h \in H,\ h \neq 1} \chi(h) = \sum_{h \in H} \chi(h) - 1 = -1,$$

because $\chi|_H$ is nontrivial.

We shall use the primitive-element indicator

$$\mathbf{1}_{\mathrm{ord}(x)=M} = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} \sum_{\mathrm{ord}(\chi)=\delta} \chi(x). \tag{S.A.1}$$

To see this, fix a generator $g$ of $L^*$ and write $x = g^a$. The left side is the indicator of $\gcd(a, M) = 1$. For a character with $\chi(g) = e^{2\pi i j/M}$ and order $\delta = M/\gcd(M, j)$, its unnormalized Fourier coefficient is the Ramanujan sum $c_M(j) = \mu(\delta)\varphi(M)/\varphi(\delta)$; Fourier inversion gives (S.A.1). (The displayed identity for $c_M(j)$ follows by Möbius inversion.)

Assume for contradiction that $E_1$ has no element of order $M$. Summing (S.A.1) over $E_1$ gives

$$0 = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} S_\delta, \qquad S_\delta = \sum_{\mathrm{ord}(\chi)=\delta} S(\chi).$$

Let $s = \omega(M)$ and $s' = \omega(q+1)$ be the numbers of distinct prime divisors. The $\delta = 1$ term is $q$. If $\delta \mid q+1$ and $\delta > 1$, then each character of order $\delta$ satisfies $\chi^{q+1} = 1$, so the corresponding contribution is $-\mu(\delta)$; summing over $\delta \mid q+1$, $\delta > 1$, gives 1. For the remaining squarefree divisors $\delta \nmid q+1$, the character sums are bounded by $\sqrt{q}$, and their number is at most $2^s - 2^{s'}$. Consequently,

$$q + 1 \leq (2^s - 2^{s'})\sqrt{q}. \tag{S.A.2}$$

Since $q+1>\sqrt{q}$, this implies

$$2^s > 2^{k/2}, \qquad \text{so} \qquad s > k/2.$$

In particular $k < 2s$. Let $p_1 < p_2 < \cdots$ be the primes and $P_s = p_1 \cdots p_s$. Since $M = 2^{2k}-1$ has exactly $s$ distinct prime divisors, $M \geq P_s$. We have

$$P_{16} = 32589158477190044730 > 2^{64}.$$

For $n \geq 16$, the ratio

$$\frac{P_{n+1}/2^{4(n+1)}}{P_n/2^{4n}} = \frac{p_{n+1}}{16}$$

is greater than 1, because $p_{n+1} \geq 59$. Hence $P_s > 2^{4s}$ for every $s \geq 16$. But $k < 2s$ gives

$$M = 2^{2k} - 1 < 2^{2k} < 2^{4s},$$

so $s \geq 16$ is impossible. Thus $s \leq 15$ and $k < 2s \leq 30$, i.e. $k \leq 29$.

It remains to cover this finite range. Each row below specifies an irreducible degree-$k$ polynomial $m_k$ over $\mathbb{F}_2$ and an element $a_k \in K_0$ with $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, where $K_0 = \mathbb{F}_2[x]/(m_k)$. The displayed hexadecimal integers encode their binary coefficient vectors. For each row let $t$ be a root of $X^2 + X + a_k$ in $K_0[t]/(X^2 + X + a_k)$. The deterministic verifier checks that $m_k$ is irreducible and that $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$. The trace condition implies that $X^2 + X + a_k$ has no root in $K_0$ and is irreducible there; the quotient is therefore a quadratic field extension. The verifier then checks that $t^{2^k} = t + 1$, and that $t^M = 1$ while $t^{M/p} \neq 1$ for every prime $p \mid M$. The last two checks show that $\mathrm{ord}(t) = M$; the third shows that $\mathrm{Tr}_{L/K_0}(t) = 1$. A verifier for the table, `verify_k1_29.py`, is included as an ancillary file. These checks are performed row by row using polynomial arithmetic in characteristic two. Its SHA-256 hash is `042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F`.

| $k$ | $m_k$ | $a_k$ |
|---|---|---|
| 1 | 0x3 | 0x1 |
| 2 | 0x7 | 0x3 |
| 3 | 0xb | 0x5 |
| 4 | 0x13 | 0xd |
| 5 | 0x25 | 0x13 |
| 6 | 0x43 | 0x34 |
| 7 | 0x83 | 0x47 |
| 8 | 0x11b | 0xc8 |
| 9 | 0x203 | 0x2b |
| 10 | 0x409 | 0xcb |
| 11 | 0x805 | 0x525 |
| 12 | 0x1009 | 0x39e |
| 13 | 0x201b | 0x1ec4 |
| 14 | 0x4021 | 0x3717 |
| 15 | 0x8003 | 0x62bf |
| 16 | 0x1002b | 0x1e6e |
| 17 | 0x20009 | 0x1155b |
| 18 | 0x40009 | 0x2ee84 |
| 19 | 0x80027 | 0xb9af |
| 20 | 0x100009 | 0x6aee2 |
| 21 | 0x200005 | 0x15750b |
| 22 | 0x400003 | 0x24a150 |
| 23 | 0x800021 | 0x6c92f9 |
| 24 | 0x100001b | 0x389ec4 |
| 25 | 0x2000009 | 0x6801ad |
| 26 | 0x400001b | 0x2707d2e |
| 27 | 0x8000027 | 0x6e9ab1e |
| 28 | 0x10000003 | 0xfac7cd2 |
| 29 | 0x20000005 | 0xc69b55e |

The table therefore supplies the required element for every $k \leq 29$. Combining this with the contradiction for $k \geq 30$ proves the theorem. □

## S.B Verification scope and original coordinate certificates

This appendix separates current checks from the archived verification record of the original 52-page manuscript. Current project paths refer to the accompanying restructured project; paths in the archived subsection refer to the original paper directory under `provenance/original_repository/`.

### Current trace certificates and Lean scope

The uniform relative-trace estimate, complete $C_9/C_{11}$ classification, variance formula and norm transfer are in main Section 5. Main Appendices B–C specify their exact-arithmetic checks. The current finite classification proof uses 755 representative witnesses and 731 support-separation witnesses; 255 further representatives at the analytic case $s = 8$ and the old 3319 all-trace families are retained as checks. These computations and the ordinary geometric proof have separate roles.

The nine current modules in `formalization/` have been checked together with Lean 4.31.0: 129 theorem and lemma declarations, plus 33 definitions. The original seven source modules retain their previous bytes. The ten grid and cover lemmas still import only `Init` and have no axiom dependencies. All remaining audits use only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`; no proof placeholder or new mathematical axiom is introduced.

The existing modules construct the multiplicative exponent equivalence, the genuine finite-field root subgroup, inverse-support closure and exact two-preimage parameter counts. The new rational transfer module constructs an actual equivalence between the admissible $\gamma$ and $a$ sets, with forward map $c/(1+\gamma)$ and inverse $(a+c)/a$. It checks the Frobenius and power identities, derives the required coprimality relations from $t = 2^s$, and proves that order $d = (t+1)(t^2+1)$ is equivalent to orders $t+1$ and $t^2+1$ of the recovered components. Its finite-set bijection transfers the primitive-element count to the filtered parameter count, and then to twice the filtered support count.

The named norm-and-trace module works with actual finite fields $K \subset E \subset L$, $|K| = 2^s$, and $[E : K] = [L : E] = 2$. It derives the quartic degree and the finite-field norm and trace power formulas from the library's `Algebra.norm` and `Algebra.trace`. The resulting population is therefore the original $M_c$, not merely a $\gamma$ count. For nonzero $c \in E$ it proves the exact counting transfer, and for $h \in K^*$, $N_{E/K}(u) = 1$, it proves the shifted support formula for $M_{hu}$. The zero slice $M_0 = 0$ and the equivalence between zero population and empty filtered support are also checked. These statements cover the manuscript's parameter range and add no new novelty claim.

Concrete field representations, the original certificate generator and its coordinate change, finite witnesses and exceptional polynomial checks remain outside Lean. The complete one-prime and prime-kernel branches, quotient-fiber counts, unique trace-class decomposition, geometric estimates, sieve and infinite-tail arguments, and the final all-parameter assembly also remain unformalized. In particular neither the complete quantitative Theorem 5.1 nor Theorem 5.4 is claimed formalized. All source hashes, declaration audits and dependency commits are in `formalization/interfaces_check.json`; `formalization/INTERFACE_SCOPE.txt` gives the detailed boundary.

The three original Lean files have also been rerun on the distributed sources in an isolated environment using the original Lean version and all nine pinned dependency commits. All three checks exit successfully. The mathlib archive cache was reused; a cache-free cloud build is not claimed. The historical `checked_environment.json` contains different hashes for `AGL.lean` and `CohenEvenConditional.lean`; it is preserved as history. The new results and the actual distributed hashes are recorded separately in `verification/supplement_sync_2026-10-26/original_lean_fresh_checks.json`. The old hash record is not used as proof that the current bytes were checked. No source file in the original Lean archive was modified to obtain the new checks.

## Archived verification record of the original manuscript

The following record is retained from the 52-page source at the pinned revision in main Appendix C. Its original run counts and formalization claims describe that source record; current scope is given above.

All numerical claims (fixed points, cycle types, group order 4032, and the characteristic polynomials of $L_T$, $L_D$, and $L_K$) were verified computationally. The script `verify_paperA_structure.py` reproduces the coordinate, cycle, group-order, basis-ordering, and $\tau$-collapse data, as well as the $\mathbb{F}_{64}$ instance of Proposition S3.2 and the CRT exponent convention. It computes the characteristic polynomials as $\det(tI + L)$ by the permutation expansion in characteristic two. The separate `verify_k1_29.py` checks the finite PTE certificate. The script `verify_complete_mapping_orbits.py` enumerates all binary matrices and translations for $n \leq 4$, checks Lemmas 2.1–2.2 and the coordinate-permutation identities of Proposition S3.1, and compares the resulting counts with Proposition S3.4. For $n = 4$ it enumerates line spreads by exact cover, orients every line, checks $F(f) = f$, and tests additivity, reproducing the counts following Proposition S3.3. The additional `verify_affine_groups_and_conjugacy.py` partitions all admissible matrices for $n \leq 4$ into linear-conjugacy classes and directly closes the two-generator permutation group for every translation at each matrix representative. The $4, 16, 80$ closure cases for $n = 2, 3, 4$, respectively,

cover all $8, 384, 93184$ affine pairs by linear conjugacy. These closures are compared with Theorem 2.3; transitivity, block systems, and double transitivity are also tested directly. The same script checks Theorem S3.7 using all 1792 maps and all 20160 linear changes of coordinates, and verifies the signed Burnside count. It shares basic permutation and spread routines with `verify_complete_mapping_orbits.py`; the direct closure computation does not use the proposed group formula. Its data file `affine_groups_and_conjugacy_results.json` includes every matrix representative and translation case, map representatives, stabilizer matrices, and the fixed-orientation counts. These deterministic scripts require only the Python standard library and were run with Python 3.12.14. The ancillary data include the 64 rows of $T, D, K$, all 720 basis orderings, the 29 PTE certificates with their factorizations, the 56 four-dimensional spreads, and a nonlinear fixed-map witness. The script `export_paperA_data.py` regenerates the coordinate and certificate tables; it reuses the two original verifiers rather than constituting an independent implementation. The scripts, data, proof sources, reproduction instructions, and the current SHA-256 manifest accompany this manuscript as ancillary files. The single command `pythonreproduce_paperA.py` checks the manifest, runs all finite verifiers, regenerates the data, and checks their hashes again. The project repository is

`https://github.com/63331849121/yijing-mathematical-structures`.

Relative to the standard basis $e_1, \ldots, e_6$, the linear part of the main map is

$$L_T = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

*Formalization.* `CompleteMappingTriality.lean` checks the function-level rotation, $F^2(f) = \mathrm{id} \oplus f^{-1}$, $F^3(f) = f$, and the unique-fixed-point assertion of Lemmas 2.1–2.2 for additive groups of exponent two.

The affine group-generation argument (for a field $\mathbb{K}$ with a primitive element $\alpha$ and $c \neq 0$, the maps $T$ and $D$ generate $\mathrm{AGL}(1, \mathbb{K})$, the core of Theorem S1.4 and §S2) and the algebraic lemmas used in it (the commutator is a translation; conjugation of translations by $T$) have been formalized and machine-checked in `AGL.lean`.

The conditional file `CohenEvenConditional.lean` formalizes the field-level Frobenius identities and the resulting order-theoretic $\tau$-collapse under an explicit order interface; it does not formalize the paper proof of the existence theorem. The concrete cycle-type data of §S1.1–S1.2 were verified by independent computation. The existence part of Theorem S2.3 is proved in Appendix S.A; its analytic part is not formalized in Lean, and its finite range is discharged by the deterministic certificate there. Thus Cohen's theorem is not used as a proof dependency; the classical references are cited only for historical attribution. The general matrix-group structure, its action criteria, the signed Burnside formula, four-dimensional conjugacy classification, coordinate-permutation description, line-spread classification, and affine orbit-count proof are paper proofs, not Lean proofs. The inverse-realization and arithmetic-sieve criteria, the primitive-group nonrealizability theorem, the positive Jacobi-sum/sieve bound, the fixed-index realization theorem and index-three threshold, the quantitative density and growing-index results, norm descent and the quarter-power obstruction, primitive inverse classification and the generating-triangle correspondence, the local invariant-orientation criterion, and the long-cycle double-transitivity criterion are also paper proofs, not part of the Lean formalization.

All three Lean files were checked using Lean 4.31.0 and mathlib commit `fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`. The ancillary `lean/` directory contains these files, `lean-toolchain`, `lakefile.lean`, and `lake-manifest.json`, pinning the proof environment. These file checks used an existing dependency cache; a fresh dependency download and cloud CI run are not claimed. Code is supplied under the MIT license, and the manuscript and data under CC BY 4.0.

The verifier `verify_standard_spread_orbits.py` independently enumerates the 60480 projective matrices for the standard six-dimensional spread and traverses all $2^{21}$ orientations using elementary matrix generators and the signed Frobenius action. It checks the projective cycle table, the weight polynomial, and the 80 disjoint orientation orbits. The data `standard_spread_results.json` contain every representative, its map, full linear stabilizer order, and both its orientation-orbit size and its full $\mathrm{GL}(6,2)$ conjugacy-class size. Only this underlying spread type is covered. The verifier `verify_nonlinear_companions.py` checks the explicit sixteen-point tables, all three cycle types, the three-cycle word certificate, and the translated-fixed identities. It does not enumerate $A_{16}$: the group equality has the elementary proof in Proposition S4.2. The arbitrary-dimensional asymptotic and translated-fixed results are paper proofs, outside the supplied Lean formalization.

The verifier `verify_scalar_realization.py` compares two independent finite counting methods: binary-polynomial gcds and explicit multiplicative subgroups in finite-field coordinates. It covers every divisor of $2^n-1$ for $n = 2, 4, 6, 8, 10, 12, 14$, checks the Jacobi bounds and slope-generation sieve, and verifies the degree-eight witness in Corollary S5.1. The data `scalar_realization_results.json` include the index-three counts $|\mathcal{R}_d| = 0, 0, 24, 90, 396, 1722$ for $n = 4, 6, 8, 10, 12, 14$, respectively. These computations verify finite arithmetic certificates; the infinite-family and threshold assertions use the paper proofs in main Section 6 and Supplementary Corollary S5.1.

The verifier `verify_scalar_depth.py` audits the quantitative main term and error by exact rational arithmetic in even dimensions 2 through 14. It checks norm descent by polynomial gcds for $2 \le r \le 8$, all proper divisors $s \mid r$, and all $u \mid 2^s - 1$. The quarter-power counts are checked for $s = 1, 2, 3, 4, 5$ by polynomial gcds and the divisor sieve, with independent field enumeration and subfield-location checks for $s = 1, 2, 3, 4$. For primitive subgroup orders in even dimensions 2 through 14, it checks Frobenius and six-ratio orbits, the degree of the polynomial encoding the realizing slopes, and the conjugacy-class counts. It also directly enumerates generating triangles and their affine stabilizers on 4 and 16 vertices. The corresponding data are `scalar_depth_results.json`. These are finite arithmetic audits; the uniform exponent obstruction and classification use the paper proofs.

### Bases for the coordinate presentation

For the coordinate presentation $T(x) = (2^{2k} - 1) \oplus \mathrm{coords}_G(x)$ on $2k$ bits the following bases give the stated group order $\langle T, D\rangle$ and $K$-cycle type:

| $k$ | basis $G$ | $\langle T, D\rangle$ | $K$ type | factor |
|---|---|---|---|---|
| 1 | $(1, 3)$ | $12 = 4 \cdot 3$ | $1 + 1 \cdot 3$ | $\Phi_3$ |
| 2 | $(1, 2, 5, 13)$ | $240 = 16 \cdot 15$ | $1 + 3 \cdot 5$ | $\Phi_5$ |
| 3 | $(63, 33, 18, 49, 1, 21)$ | $4032 = 64 \cdot 63$ | $1 + 7 \cdot 9$ | $\Phi_9$ |
| 4 | $(134, 136, 241, 71, 34, 65, 3, 143)$ | $65280 = 256 \cdot 255$ | $1 + 15 \cdot 17$ | a degree-8 factor of $\Phi_{17}$ |

All four rows were verified by independent computation. The 720-case count was reproduced by enumerating all orderings of the six displayed vectors, solving the corresponding $\mathbb{F}_2$ linear system

for coords$_G$, and recording the cycle types of $T, D, K$ whenever the two inverses exist.

### Explicit tau-collapse data

With $\mathbb{F}_{2^{2k}} = \mathbb{F}_2[t]/(f)$ and elements written as polynomials in $t$, the following rows realize the order conditions of Theorem S2.3:

| $k$ | $f$ | $\gamma$ | $\tau = \gamma \oplus \gamma^{-1}$ | $\alpha = (1 \oplus \gamma)^{-1}$ |
|---|---|---|---|---|
| 1 | $t^2 + t + 1$ | $t$ | $1$ | $t$ |
| 2 | $t^4 + t + 1$ | $t^3$ | $t^2 + t + 1$ | $t$ |
| 3 | $t^6 + t + 1$ | $t^2 + t$ | $t^4 + t^3 + 1$ | $t^5 + t^3 + t^2$ |
| 4 | $t^8 + t^4 + t^3 + t^2 + 1$ | $t^3 + t^2 + t + 1$ | $t^7 + t^4 + t^3 + 1$ | $t^6 + t^4 + t^3 + t^2 + 1$ |

In each row $\mathrm{ord}(\gamma) = \mathrm{ord}(1 \oplus \alpha^{-1}) = 2^k + 1$, $\tau$ is primitive in $\mathbb{F}_{2^k}$, and $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = 2^{2k} - 1$ (all verified). The defining polynomial $f$ is a field-representation choice and need not coincide with the characteristic polynomial $p_T$ of (S1.1); for instance, the $k = 3$ row uses $f = t^6 + t + 1$, while $p_T(t) = t^6 + t^5 + t^3 + t^2 + 1$.

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